

Progress Report 1 (Updated)

BAN's Protocol: Adaptive Horde Survival

A fuzzy-logic-driven survival shooter (HTML5 Canvas + JavaScript)

Course: ISP568 — Fuzzy Logic Systems · **Coursework Weight:** 15%

What changed since the first submission. The original report described a single `bans-protocol.html` file with **3 fuzzy inputs** (Health, Ammo, Noise) and an **11-rule** hand-written base. Through four development phases (see §1.0) the prototype was rebuilt into a **modular 17-file codebase** and the fuzzy controller grew into a complete **6-input → 3-output → 729-rule (3⁶) Mamdani MIMO system**: inputs **Health, Ammo, Noise, Pressure, Exposure, Skill**; outputs **Threat, Supply, Composition**. Around it we added a procedurally generated maze with line-of-sight / fog stealth, perception-based enemy AI, a per-enemy micro-fuzzy controller, a roguelite progression layer (perks, sector modifiers, deployables, achievements), a full UX layer (settings, pause, tutorial, gamepad/touch), an original procedural + streamed soundtrack, and an in-game **Fuzzy Analytics** tool (control-surface plot, rule heatmap, defuzzifier comparison, CSV export). This report supersedes the previous one.

1.0 Development Phases — How the Game Evolved

Development ran in four phases on top of the original prototype. Each phase was implemented, verified in a browser, and documented before the next began.

Phase	Theme	Key additions
Base → v5	Core fuzzy director + world	Rebuild to modular JS · 3→5 fuzzy inputs (added Pressure, Exposure) · 11→243 generated rules · procedural maze, LOS/fog stealth, perception AI · 2nd output Supply · per-enemy micro-FIS · weapons/coins/power-ups/upgrades · procedural audio + Molchat-Doma soundtrack + beat-reactive lighting · 2.5D neon visuals, decals, dust, weather · boss waves + roguelite sectors (maze reshuffles each boss)
Phase 1	Combat depth	Spitter (ranged) enemy + enemy projectiles · elite affixes (shielded / volatile / frenzied) · boss attack patterns (radial burst + charge) · dodge dash (i-frames) · melee knife + silent stealth takedowns · combo/killstreak multiplier · damage-direction indicator · boss minimap ping
Phase 2	Roguelite variety	Pick-1-of-3 perks after each boss (12 run modifiers) · random sector modifiers · deployable mine + fuzzy turret · exploding barrels + destructible walls · extraction-beacon objective · 11 achievements · daily-seed challenge
Phase 4	UX & polish	Settings menu (volume / toggles) · pause menu + stats · first-run tutorial · threat-dynamic music · player-cast wall shadows · gib death effects · gamepad + touch · offscreen floor cache · central balance config
Phase 3	Fuzzy depth (this report)	6th input Skill (adaptive difficulty) · 3rd output Composition (enemy-type mix) · 243 → 729 rules · Takagi–Sugeno path + runtime Mamdani↔Sugeno toggle · in-game Fuzzy Analytics : control-surface plot, rule-firing heatmap, defuzzifier comparison, "why this threat?" explainability, fuzzy weapon advisor, CSV data-logger

1. App / Game Prototype Generation

1.1 Concept & Idea

BAN's Protocol: Adaptive Horde Survival is a top-down survival shooter set inside a randomly generated maze. The enemy horde is not driven by random spawns or a fixed difficulty curve — a **fuzzy-logic "AI director"** continuously reads the player's live state and decides how aggressive the horde should be and how many enemies to send.

The defining idea added since the last report is **stealth-aware difficulty**: the director now cares not only about *how strong* the player is, but about **whether the horde actually knows where the player is**. A player who hides in fog and stays quiet is left alone almost regardless of how well-armed they are; a player who is seen, loud and loaded gets the

full swarm. Difficulty has become a resource the player manages through *positioning and noise discipline*, not just through health and ammo.

1.2 Problem Statement

Conventional games scale difficulty with crisp thresholds or fixed timers (e.g. "after 60 seconds, spawn 10 enemies"). These rules are brittle: they snap between levels, ignore the player's actual situation, and cause either boredom (too easy) or a frustrating death spiral (piling enemies onto an already-weak player). The problem we address is **dynamic difficulty balancing under multiple, overlapping, vague conditions**: how can a game smoothly and believably adapt its challenge to the player's moment-to-moment condition — health, firepower, noise, how swarmed they already are, and how exposed they are — when every one of those signals is naturally fuzzy ("fairly low health", "kind of loud", "mostly hidden")?

1.3 Target Users

- **Casual and mid-core action / survival players** who want a challenge that always feels fair.
- **Game designers and students** studying adaptive / dynamic-difficulty AI directors.
- **Educators and learners** using the live on-screen fuzzy visualisation as an interactive demonstration of Mamdani fuzzy inference.

1.4 Prototype — UI Layout & Interaction Flow

The prototype is a **complete, playable build** in pure HTML5 Canvas + JavaScript. Every control and readout is drawn on the canvas (no HTML form widgets). The interaction flow is:

- **Mission Setup screen**: canvas-drawn draggable sliders — **Game Pace, Horde Density, Survivability** — a **God-mode toggle** for demos, and a **DEPLOY** button.
- **ARMORY (shop)**: buy/upgrade/equip the three weapons and four permanent operative upgrades using coins earned in previous runs.
- **Gameplay**: **WASD/Arrows** move, mouse aims, **Click/Space** shoots, **R** reloads, **Q / 1 / 2 / 3** switch weapons, **G** toggles god mode, **B / V** toggle HUD panels, **Esc** pauses.
- **Live HUD**: operative vitals (Health, Ammo, Noise), a circular **Threat / Aggression gauge**, mission stats and coins, a **minimap** (walls, fog, player and every enemy as a red dot), **five live membership-function graphs**, and an **Active Fuzzy Rules** panel.
- **Game Over (OVERRUN) screen** with score/kills/wave and a **REDEPLOY** button; coins are banked to local storage on death so they persist between runs.

1.5 Core Functional Components

All core components are implemented and working (not placeholders):

- **Fuzzy inference engine** (`js/fuzzy.js`) — triangular/trapezoidal membership functions, a **complete 729-rule** Mamdani base (AND = min, aggregation = max), centroid defuzzification. The module is pure and game-agnostic: it takes `{health, ammo, noise, pressure, exposure}` in 0–100 and returns a crisp Threat plus all inference internals.
- **Procedural maze world** (`js/world.js`) — a 39×23-tile maze (3900×2300 px) carved by recursive backtracking, then *braided* (loops opened) with a few open plazas, regenerated every deploy; wall collision, **line-of-sight sampling**, **fog hiding zones**, and a prerendered minimap.
- **Real-time game loop** (`js/mechanics.js`) — player movement with wall-sliding, aiming, shooting, reloading, explosions, pickups, and the situation-aware spawn director.
- **Perception-based enemy AI** — enemies **chase** only with line-of-sight (blocked by walls and fog), **hunt** toward noise with positional error that shrinks as the player gets louder, and otherwise **wander**; three visually distinct

archetypes (grunt / runner / brute).

- **Live fuzzy visualisation** (`js/hud.js`) — threat gauge with needle, five MF graphs with a moving marker per input, and an active-rules panel listing which rules fire and at what strength.
- **Progression systems** (`js/weapons.js`) — coins, three weapons with +25%/level upgrades, six power-ups, four permanent operative upgrades, all persisted to `localStorage` .

1.6 Rationale — Why Fuzzy Logic Is Appropriate

The director's decision depends on human, imprecise concepts: "a bit low on health", "fairly loud", "almost out of ammo", "kind of swarmed", "barely spotted". These categories overlap and have no sharp boundary — exactly the uncertainty fuzzy logic is designed for. Concretely:

- **Smooth blending** — overlapping membership functions interpolate between rules, so difficulty ramps and eases gradually instead of snapping between levels.
- **Natural rule authoring** — rules read in plain language ("IF health is Low AND ammo is Depleted THEN threat is Passive").
- **Multi-input fusion** — *five* uncertain inputs are combined into one coherent decision. With five variables, hand-coding every combination is infeasible; fuzzy inference fuses them cleanly (and we generate the full 729-rule grid from three transparent scoring policies — see §2.4).
- **Believable behaviour** — the resulting difficulty curve feels reactive and intentional rather than scripted, and the stealth dimension (Exposure) gives the player real agency over the challenge.

1.7 Section Categories — the Subsystems Inside the Game

The codebase is organised into clearly separable subsystem categories. Each maps to one or more source files (full file-by-file guide in Appendix A):

Category	What it does	Main files
Fuzzy AI (core)	The 729-rule macro director, per-enemy micro-FIS, weapon-advisor FIS, and the analytics/inspection tools	<code>fuzzy.js</code> , <code>analytics.js</code>
World	Procedural maze, collision, line-of-sight, fog, destructible walls, minimap, daily seed	<code>world.js</code>
Simulation / director	Game loop, movement, combat, enemy AI, spawn director, deployables	<code>mechanics.js</code>
State	The single <code>G</code> state object + <code>reset()</code>	<code>state.js</code>
Combat & progression	Weapons, levelling, power-ups, coins, roguelite perks / sectors / achievements / daily	<code>weapons.js</code> , <code>roguelite.js</code>
Rendering & FX	World/character rendering, lights, decals, dust, weather, bloom, tuning config	<code>render.js</code> , <code>post.js</code> , <code>balance.js</code>
HUD & UI	Gauges, MF graphs, live rules, minimap; setup / shop / pause / death / settings / perk / tutorial screens	<code>hud.js</code> , <code>screens.js</code>
Audio	Procedural SFX, streamed soundtrack, adaptive/threat-dynamic music, beat-reactive lighting	<code>audio.js</code>
Input & loop	Keyboard / mouse / gamepad / touch; the requestAnimationFrame driver	<code>input.js</code> , <code>main.js</code> , <code>config.js</code>

2. Identification of Fuzzy Inputs

2.1 Proposed Fuzzy Input Variables

Five crisp game-state values are normalised to a 0–100 universe of discourse and fuzzified. The Mamdani controller now defuzzifies **three** outputs from those same six inputs — **Threat** (how hard the horde pushes) and the newly added **Supply** (how generous the director's pickup drops are) — making it a multiple-input, multiple-output (MIMO) fuzzy system.

Variable	Type	Universe (0–100)	Linguistic terms	Justification
Player Health	Input	% of max HP	Low, Medium, High	How close the player is to death; the baseline signal of how much pressure is fair.
Ammo Count	Input	% of total capacity	Depleted, Adequate, Surplus	The player's ability to fight back; low ammo should not be punished with a swarm.
Noise Level	Input	accumulates on firing/moving, decays over time	Quiet, Moderate, Loud	How much attention the player draws; rewards stealthy play and explains <i>why</i> the horde grows.
Pressure (added v2)	Input	nearest-enemy distance + crowd count	Safe, Engaged, Swarmed	How swarmed the player <i>already</i> is — a relief valve so the director does not pile onto a buried player.
Exposure (added v3)	Input	detection meter (line-of-sight, fog, noise)	Hidden, Suspicious, Spotted	Whether the horde actually <i>knows where the player is</i> ; the dominant input — an unseen player is left alone.
Skill (added Phase 3)	Input	rolling performance estimate (accuracy, kill-rate, depth, damage taken)	Novice, Average, Expert	How good the player <i>is</i> . Adaptive difficulty: expert players are pushed harder, novices eased. Centred on Average, so an average player sees the original balance.
Threat / Aggression	Output 1	enemy intensity	Passive, Tactical, Overwhelming	Defuzzified value that sets horde size, spawn cadence, spawn position, enemy speed, damage and colour.
Supply / Director Aid (added v4)	Output 2	drop generosity	Scarce, Balanced, Generous	Defuzzified value that scales how often kills drop health / ammo / power-ups — generous to the desperate, stingy to the comfortable, decoupled from Threat.
Composition (added Phase 3)	Output 3	enemy-type mix	Swarm, Mixed, Heavy	Defuzzified value biasing <i>what kind</i> of horde spawns — cheap grunts (Swarm) vs. brutes / spitters / elites (Heavy). Decides horde <i>quality</i> , separately from Threat's <i>quantity</i> .

How the two new crisp inputs are computed (in `js/mechanics.js`, every 10 frames):

```
// Pressure: closer + more crowded ⇒ higher
const press = G.enemies.length===0 ? 0
               : Math.max(0, Math.min(100, (100 - nd/5) + near*2 )); // nd = nearest-enemy distance

// Exposure (G.detect): line-of-sight ramps it up, fog drains it, loud noise sets a floor
if(seen)      G.detect = Math.min(100, G.detect + 9);           // any enemy with LOS
else          G.detect = Math.max(0, G.detect - (inFog ? 2.5 : 0.5)); // fog hides ~5× faster
if(G.noise>45) G.detect = Math.max(G.detect, Math.min(65, G.noise*0.65)); // "suspicious" floor
```

2.2 Crisp → Linguistic Mapping (Fuzzy Graphs)

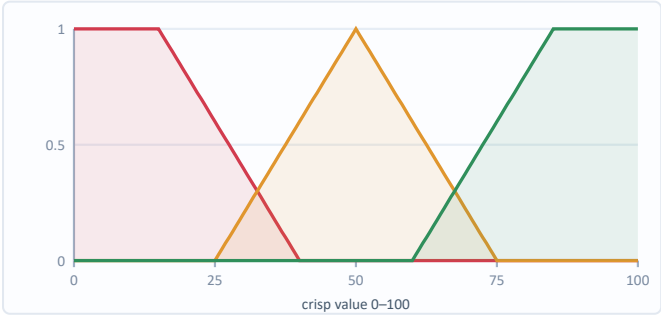
Each input is mapped from a crisp number to degrees of membership in its terms. In the running prototype these graphs are drawn live on the canvas, one per input, with a moving marker crossing each curve. The exact shapes implemented in `js/fuzzy.js` are:

Variable	Term	MF shape	Defining points (0–100)
Player Health	Low	Left shoulder	1 below 15, ramps to 0 at 40
Player Health	Medium	Triangle	0 at 25, peak 50, 0 at 75
Player Health	High	Right shoulder	0 at 60, ramps to 1 at 85
Ammo Count	Depleted	Left shoulder	1 below 10, ramps to 0 at 35
Ammo Count	Adequate	Triangle	0 at 25, peak 55, 0 at 80
Ammo Count	Surplus	Right shoulder	0 at 65, ramps to 1 at 90
Noise Level	Quiet	Left shoulder	1 below 15, ramps to 0 at 40
Noise Level	Moderate	Triangle	0 at 30, peak 55, 0 at 80
Noise Level	Loud	Right shoulder	0 at 65, ramps to 1 at 90
Pressure	Safe	Left shoulder	1 below 20, ramps to 0 at 45
Pressure	Engaged	Triangle	0 at 35, peak 60, 0 at 85
Pressure	Swarmed	Right shoulder	0 at 70, ramps to 1 at 90
Exposure	Hidden	Left shoulder	1 below 15, ramps to 0 at 40
Exposure	Suspicious	Triangle	0 at 30, peak 55, 0 at 80
Exposure	Spotted	Right shoulder	0 at 65, ramps to 1 at 90
Threat (output 1)	Passive	Triangle	0 at 0, peak 18, 0 at 40
Threat (output 1)	Tactical	Triangle	0 at 30, peak 50, 0 at 70
Threat (output 1)	Overwhelming	Right shoulder	0 at 60, ramps to 1 at 90
Supply (output 2)	Scarce	Triangle	0 at 0, peak 18, 0 at 40
Supply (output 2)	Balanced	Triangle	0 at 30, peak 50, 0 at 70
Supply (output 2)	Generous	Right shoulder	0 at 60, ramps to 1 at 90

The same mappings plotted as **membership-function graphs** — the six inputs (Skill is the new 6th) followed by the three outputs (Composition is the new 3rd). The x-axis is the crisp 0–100 value, the y-axis is the degree of membership $\mu \in 0\text{--}1$; overlapping curves are what let the engine blend smoothly between rules:

Player Health (input 1)

■ Low ■ Medium ■ High



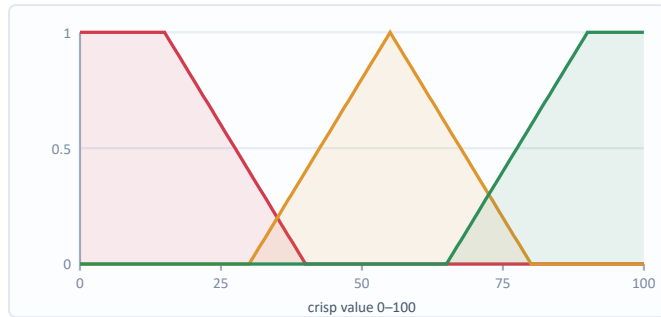
Ammo Count (input 2)

■ Depleted ■ Adequate ■ Surplus



Noise Level (input 3)

■ Quiet ■ Moderate ■ Loud



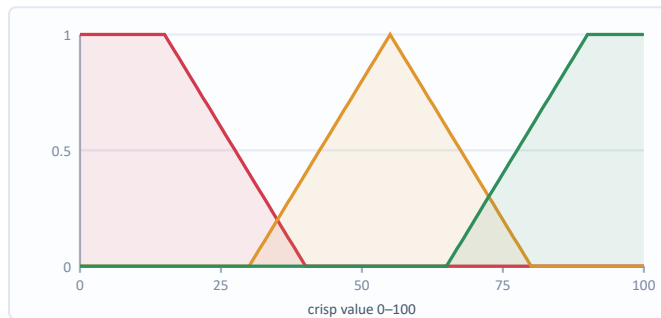
Pressure (input 4)

■ Safe ■ Engaged ■ Swarmed



Exposure (input 5)

■ Hidden ■ Suspicious ■ Spotted



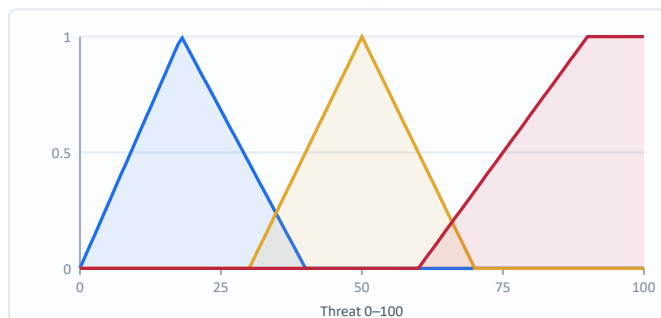
Player Skill (input 6 — NEW)

■ Novice ■ Average ■ Expert



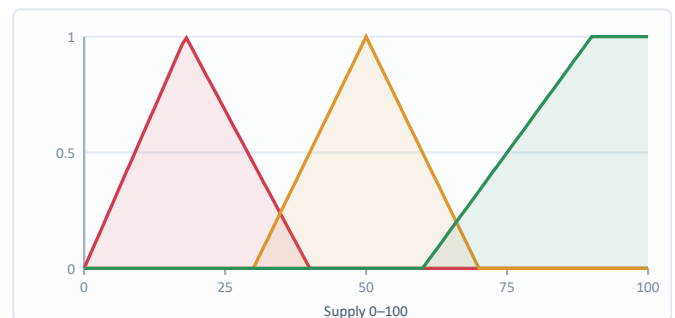
Threat / Aggression (OUTPUT 1)

■ Passive ■ Tactical ■ Overwhelming



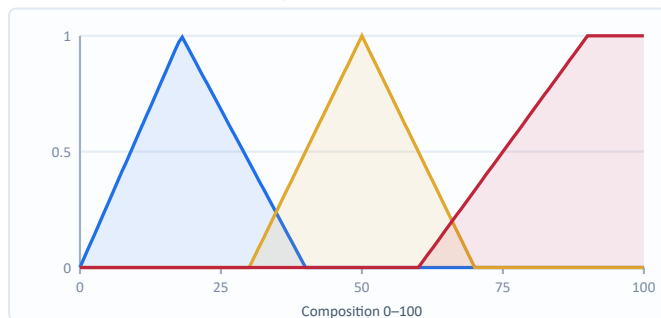
Supply / Director Aid (OUTPUT 2)

■ Scarce ■ Balanced ■ Generous



Composition (OUTPUT 3 — NEW: enemy-type mix)

■ Swarm ■ Mixed ■ Heavy



2.3 Membership Function Ideas (not yet finalised)

- Triangular and trapezoidal functions are kept for speed and easy real-time plotting; Gaussian curves may be trialled later for smoother transitions.
- The overlap points (e.g. Health Low/Medium crossing near 30–40, Exposure Hidden/Suspicious near 30–40) are tuning targets to be balanced through playtesting.

- Noise gain-per-shot (12–32 depending on weapon), the $-0.35/\text{frame}$ decay, the Exposure ramp/drain rates and the noise "suspicious" floor are provisional and will be calibrated so each input meaningfully influences the output.

2.4 Expected Fuzzy Rules — the three generating policies

The inference is a **Mamdani** model with **AND = minimum**, **aggregation = maximum**, defuzzified by **centroid** (with a parallel Sugeno path, §3.8). Rather than hand-write 729 rules, the engine **generates the complete $3^6 = 729$ -rule grid**; each rule carries **three** consequents (Threat, Supply, Composition), each assigned by its own transparent policy over the ordinal term levels (0/1/2):

```
THREAT = 0.55·(health+ammo) + 0.8·noise - 0.6·pressure + 1.7·exposure + 0.6·(skill-1)
        ≤ 2.3 → Passive · ≤ 4.6 → Tactical · else → Overwhelming

SUPPLY = (2-health) + (2-ammo) + 0.6·pressure (help the desperate)
        ≤ 1.6 → Scarce · ≤ 3.0 → Balanced · else → Generous

COMPO = 0.5·(health+ammo) + 0.7·noise + 0.8·exposure + 0.5·skill (heavier when strong/loud/seen/skilled)
        ≤ 2.0 → Swarm · ≤ 3.8 → Mixed · else → Heavy

weight w = 0.6 (at a band boundary) ... 1.0 (deep inside a band)
```

- **Capability raises Threat** but is damped (0.55) — strong, loaded players get pushed.
- **Noise raises Threat** (0.8) — being loud is punished.
- **Pressure lowers Threat** (-0.6) — already-swarmed players get relief, not a pile-on.
- **Exposure dominates Threat** (1.7) — a Hidden & Quiet player stays Passive no matter how strong.
- **Skill is centred** ($0.6·(\text{skill}-1)$): Average adds 0, so an average player sees exactly the pre-Phase-3 balance; Novice is eased and Expert is pushed — adaptive difficulty without breaking the tuning.
- **Supply** ignores Noise/Exposure so it is genuinely decoupled from Threat; **Composition** rewards seen/loud/skilled play with a tougher enemy *mix*.

Rule-base composition (machine-enumerated over all 729 rules): Threat = 271 Passive · 308 Tactical · 150

Overwhelming; Supply = 189 Scarce · 270 Balanced · 270 Generous; Composition = 156 Swarm · 408 Mixed · 165

Heavy. The Threat distribution is deliberately mercy-biased — well under a third of situations unleash the full horde, reserved for players who are healthy, loaded, loud, seen **and** skilled.

Representative sample (the brief asks for ≥ 2 draft rules; **all 729 are listed exhaustively in Appendix B**). The 5 antecedents shown pin Skill = Average for readability:

#	Antecedent (IF)	Score	Consequent (THEN Threat)	Design intent
1	Health Low AND Ammo Depleted ($\wedge$ Quiet $\wedge$ Safe $\wedge$ Hidden)	0.0	Passive	Mercy: a wounded, defenceless, unseen player gets breathing room.
2	Health High AND Ammo Surplus $\wedge$ Loud $\wedge$ Safe $\wedge$ Spotted	7.2	Overwhelming	Challenge: a strong, loud, fully-armed player who is <i>seen</i> gets the hardest fight.
3	Health High $\wedge$ Ammo Surplus $\wedge$ Quiet $\wedge$ Safe $\wedge$ Hidden	2.2	Passive	Stealth override: even a strong player, if hidden and quiet, is left alone.
4	Health Medium $\wedge$ Ammo Adequate $\wedge$ Moderate $\wedge$ Engaged $\wedge$ Suspicious	~ 3.6	Tactical	Balanced, partially-spotted state gets standard, steady pressure.

#	Antecedent (IF)	Score	Consequent (THEN Threat)	Design intent
5	Health High $\wedge$ Ammo Surplus $\wedge$ Quiet $\wedge$ Swarmed $\wedge$ Spotted	~3.9	Tactical	A strong player already buried in enemies gets <i>relief</i> , not a pile-on.

2.5 Rule-Base Justification — input/output count, why 729, and direct vs combined rules

(a) **How many inputs and outputs are actually used, and why each one.** The controller is a **six-input, three-output** MIMO fuzzy model. It grew MISO (1 output) $\rightarrow$ MIMO (2, then 3) across the phases. Every variable earns its place — none is decorative:

Direction	Variable	Terms (3 each)	Why it is a fuzzy variable
IN 1	Health	Low · Medium · High	"a bit low on health" is inherently vague — the baseline of how much pressure is fair.
IN 2	Ammo	Depleted · Adequate · Surplus	ability to fight back; a dry player should not be swarmed.
IN 3	Noise	Quiet · Moderate · Loud	attention drawn — the input the player directly authors, making difficulty steerable.
IN 4	Pressure	Safe · Engaged · Swarmed	how swarmed you <i>already</i> are — a relief valve against pile-ons.
IN 5	Exposure	Hidden · Suspicious · Spotted	whether the horde <i>knows where you are</i> — the dominant, stealth-defining input.
IN 6	Skill	Novice · Average · Expert	how good the player is — turns a fixed director into a player-adaptive one.
OUT 1	Threat	Passive · Tactical · Overwhelming	how <i>hard</i> the horde pushes (quantity: size, cadence, speed, damage).
OUT 2	Supply	Scarce · Balanced · Generous	drop generosity — a rubber-band that helps the desperate, decoupled from Threat.
OUT 3	Composition	Swarm · Mixed · Heavy	what <i>kind</i> of horde (quality: grunts vs. brutes/spitters/elites), separate from quantity.

Why three separate outputs and not one. The three decisions are genuinely independent: a *hidden but hurt* player should face **low Threat** yet receive **generous Supply**; a *skilled, spotted* player warrants **high Threat** and a **Heavy** composition. Folding them into one number could not express these — so the same antecedents drive three consequent columns, each with its own scoring policy (§2.4). (Score, Kills and Wave are crisp *outcome metrics*, not fuzzy outputs — see §3.9.)

(b) **Why 729 rules.** A complete grid is *terms^{inputs}*. With **six** inputs of **three** terms, the full antecedent space is **$3^6 = 729$** , and the engine generates *all* of it so *no reachable game state is ever unhandled* — the number is a direct, unavoidable consequence of choosing six meaningful inputs, not padding. Capping lower would silently drop inputs that define the game's feel (Pressure, Exposure, Skill):

Fuzzy inputs	Terms each	Complete rule grid	Covers
3	3	$3^3 = 27$	Health, Ammo, Noise only
4	3	$3^4 = 81$	+ Pressure
5	3	$3^5 = 243$	+ Exposure (the v3–v5 build)
6	3	$3^6 = 729$	+ Skill — the full system used here

The brief's "*at least 2 rules*" is a floor, not a target; 729 is the natural, complete consequence of choosing six inputs and demanding total coverage. Crucially the rules are **not hand-written** (which would invite contradictions at this scale) — all 729 are generated from three transparent scoring policies (§2.4), so the grid stays consistent and, as this shows, **the policy approach scales to any number of inputs without extra authoring effort** — a key engineering advantage over hand-built rule bases.

Managing 729 rules for a human. Because the base is generated and every rule fires with a weight, the player never confronts 729 rules at once — typically only a handful fire meaningfully at any instant. The in-game **ACTIVE FUZZY RULES** panel lists the top-firing rules live, and the **Fuzzy Analytics** screen (§3.10) shows a **rule-firing heatmap** so the whole 729-cell base can be inspected at a glance.

(c) Direct rules or combined rules? A **direct** rule has a single antecedent — one input mapped straight to the output (e.g. the lecture example "*IF x is A1 THEN z is C3*"). A **combined** rule joins several antecedents with a fuzzy operator AND/OR (e.g. "*IF x is A2 AND y is B2 THEN z is C2*"). **Every one of our 729 rules is a combined (conjunctive) rule:** each rule fixes one term for *all six* inputs, joined by **AND = min**. This is deliberate — Threat depends on the *interaction* of the five signals (a strong player who is *hidden* must be treated completely differently from a strong player who is *spotted*), and that cross-coupling can only be expressed by combined rules. A set of independent direct rules (one per input) could never produce the "strong-but-hidden ⇒ Passive" behaviour that defines the game.

2.6 Extra Fuzzy Controllers — the Per-Enemy Micro-FIS and Weapon Advisor

Beyond the 729-rule macro director (which now carries three consequents — Threat in §3.7, Supply and Composition policies in §2.4), the game runs **two additional, independent fuzzy controllers**, showing fuzzy logic applied at several scales at once.

(For reference, the SUPPLY output was the first multi-output extension.) The Supply consequent is assigned from a second transparent scoring policy (term levels 0/1/2):

```
supplyScore = (2 - health) + (2 - ammo) + 0.6·pressure           // desperation index

consequent:  supplyScore ≤ 1.6 → Scarce      ·      ≤ 3.0 → Balanced      ·      else → Generous
```

- **Low health and low ammo raise Supply** — the desperate get more drops.
- **Being swarmed (Pressure) raises Supply** — relief for a buried player.
- It does *not* use Noise or Exposure, so Supply is genuinely **decoupled from Threat**: a hidden player can be left alone (low Threat) yet still showered with pickups if they are hurt (high Supply).

Supply composition (over all 729 rules): 189 Scarce · 270 Balanced · 270 Generous. In the game this value scales the kill-drop probability (**0.20 + 0.30·supply** for health/ammo, and the rare power-up chance), shown live as the SUPPLY bar and its membership graph.

(a) Per-enemy micro-FIS — a second, smaller fuzzy controller. While the 729-rule system is the macro "director", each enemy now runs its own tiny **3-input, 27-rule** Mamdani controller every few frames to decide *how* it approaches (its *behaviour*, not the global difficulty):

Direction	Variable	Linguistic terms (3 each)
IN 1	Distance to player	Close · Mid · Far
IN 2	Own health	Low · Med · High
IN 3	Allies nearby	Few · Some · Many
OUT	Aggression	Flee · Hold · Swarm

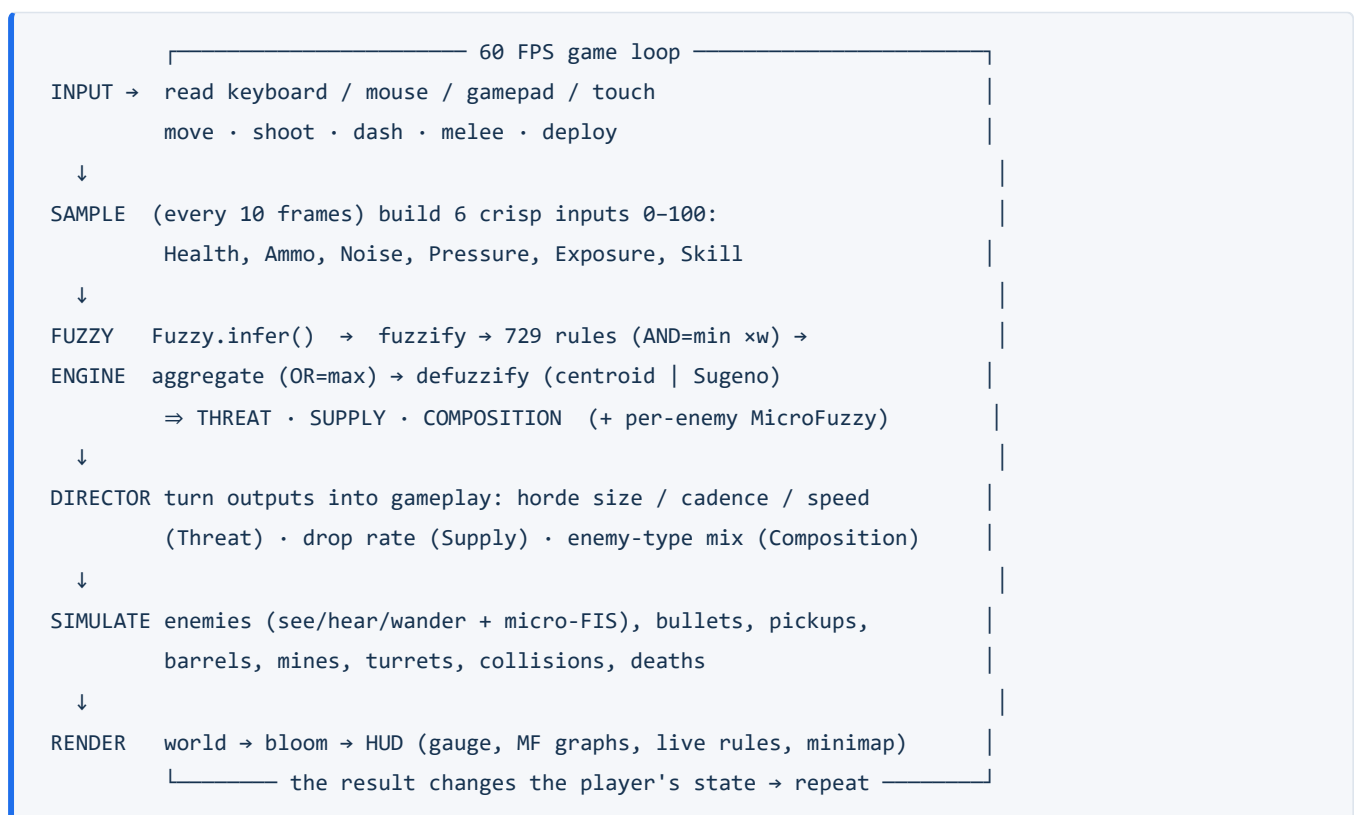
The complete antecedent space is $3^3 = 27$ rules, generated from `aggression = ownHealth + alliesNearby + 0.4·closeness` (bands: $\leq 1.5 \rightarrow \text{Flee}$, $\leq 3.0 \rightarrow \text{Hold}$, else $\rightarrow \text{Swarm}$), giving **7 Flee · 13 Hold · 7 Swarm**. Defuzzified by centroid to an Aggression 0–100, it makes a hurt, isolated creature *retreat*, a healthy creature in a pack *swarm*, and the middle *flank* — fuzzy logic operating at the micro scale as well as the macro. This demonstrates the same Mamdani pipeline (fuzzify $\rightarrow$ min $\rightarrow$ max $\rightarrow$ centroid) reused for a completely different decision.

3. Project Explanation & Fuzzy Logic Functionality

The system is a **modular codebase** (no single mega-file). `index.html` loads the modules in dependency order; the fuzzy engine lives in `js/fuzzy.js` and is queried by the game loop ~6 times/sec.

3.0 System Flow — How It All Fits Together

The game is a **closed feedback loop** ticking at 60 FPS. One pass of the loop (`frame()` in `main.js` $\rightarrow$ `update()` in `mechanics.js`) does this:



The key property is that there is **no scripted timeline**: difficulty is an *emergent equilibrium* between the player and the fuzzy director. Player actions change the six inputs $\rightarrow$ the engine re-decides the three outputs $\rightarrow$ the director reshapes the horde $\rightarrow$ the new horde changes the player's state $\rightarrow$ the loop repeats. Skill closes a second, slower loop: sustained good play raises the Skill input, which permanently nudges the whole system harder (and vice-versa). Clearing a boss triggers a *sector reshuffle* (new maze + modifier) and a perk choice, so the loop also has long-term roguelite progression layered on top.

3.1 What We Built — System Overview

Subsystems communicate through a shared game-state object `G`:

- **Fuzzy engine** (`Fuzzy` , `js/fuzzy.js`) — pure functions for membership, inference, defuzzification.
- **World** (`js/world.js`) — maze generation, wall collision, line-of-sight, fog, minimap.
- **Game state** (`G` , `js/state.js`) — player, enemies, bullets, particles, settings, latest fuzzy result.

- **Update loop** (`js/mechanics.js`) — input, movement, collisions, enemy AI, samples the player's state and asks the fuzzy engine for a Threat value.
- **Director / spawner** (`js/mechanics.js`) — turns Threat into spawning decisions.
- **Renderer + HUD** (`js/render.js` , `js/hud.js`) — world plus the fuzzy visualisations.
- **Progression** (`js/weapons.js`) — coins, weapons, power-ups, upgrades, persistence.

The key idea is a **closed feedback loop**: the player's actions change Health / Ammo / Noise / Pressure / Exposure → the fuzzy engine converts that into a Threat level → the director spawns enemies accordingly → fighting and hiding changes the player's state again → repeat. Fuzzy logic sits at the centre as the decision-maker.

3.2 The Fuzzy Pipeline (Step by Step)

Every inference pass follows the classic Mamdani sequence. Each step corresponds to real code.

Step 1 — Sampling & Normalisation. Every 10 frames the loop reads five raw values and normalises each to 0–100:

- Health = $HP / \text{maxHP} \times 100$
- Ammo = $(\text{magazine} + \text{reserve}) / \text{total capacity} \times 100$
- Noise = a running meter (+12...32 per shot by weapon, +0.5/frame while moving, -0.35/frame decay)
- Pressure = $(100 - \text{nearestDist}/5) + \text{nearbyCount} \cdot 2$, clamped 0–100
- Exposure = the detection meter `G.detect` (LOS up, fog down, loud-noise floor)

Step 2 — Fuzzification. Each crisp value passes through its term functions (`tri` , `trapL` , `trapR`) to give a membership 0–1. Overlap lets the system blend between rules instead of snapping.

Step 3 — Rule evaluation. Each of the 729 rules combines its six antecedents with AND = min, then multiplies by the designer weight `w`: $\text{strength} = \min(\mu_1, \dots, \mu_6) \cdot w$. Firing rules (strength > 0.001) are recorded for the on-screen panel.

Step 4 — Aggregation. Rules pointing at the same output term are combined with OR = max, giving one aggregated strength per output set (Passive / Tactical / Overwhelming).

Step 5 — Defuzzification. Centroid over the clipped, unioned output sets, sampled in steps of 2:

```
for x = 0..100 step 2:
    μ = max over terms of min(agg[term], term_MF(x))
    num += x·μ ; den += μ
threat = num / den
```

Centroid gives smooth, continuous output — small state changes move Threat gently, exactly what a dynamic-difficulty director needs.

3.3 From Threat Number to Gameplay

The crisp Threat (and its band) drives the director:

Parameter	How Threat affects it
Horde size per wave	<code>count = round((0.5 + aggr·4.5) · Density)</code> — higher Threat sends more enemies (enemy cap 40).
Spawn cadence	<code>spawnTimer = (240 - Threat·1.5) / Pace</code> — a strong, seen player faces wave after wave.
Spawn position	ring around the player at open maze tiles, off-screen; tighter ring at high aggression.
Enemy speed & damage	both scale with aggression; archetype mix shifts (runners when Loud, brutes when pushing).

Parameter	How Threat affects it
Enemy colour	blue (Passive) → orange (Tactical) → red (Overwhelming) for instant visual feedback.
Warm-up grace	for the first 600 frames (~10 s) the effective Threat is capped at 40 so runs start readable.

Stealth override: when detection < 15 the director **ignores Threat entirely** and spawns purely from Noise — dead silence yields at most a lone wanderer; gunfire scales the swarm with how loud you are.

3.4 The Stealth Loop — Closing It with Exposure

Noise and Exposure together give the player agency over difficulty. Firing is loud and draws line-of-sight, pushing Exposure toward *Spotted*, which fires high-threat rules. Slipping into **fog** drains detection ~5× faster (hiding resets aggression in about a second) and holding fire lets Noise decay to *Quiet*, dropping Threat. Difficulty becomes a strategic resource — cleanly expressible only because Noise and Exposure are fuzzy quantities, not hard on/off flags.

3.5 Live Visualisation of the Fuzzy System

So the inference is not a black box, the game renders it in real time: the **Threat gauge** (needle + band arc), **five MF graphs** (one per input, moving marker + dominant-term highlight), and the **Active Rules panel** (which rules fire and at what strength, as percentage bars) — an X-ray of the inference each instant. A **minimap** shows walls, fog, the player and enemies so the stealth state is legible too.

3.6 Worked Example

The player is healthy and fully loaded but **hiding in fog and not shooting**: Health ≈ 90 (High ≈ 1), Ammo ≈ 95 (Surplus ≈ 1), Noise ≈ 5 (Quiet ≈ 1), Pressure ≈ 0 (Safe ≈ 1), Exposure ≈ 5 (Hidden ≈ 1). The dominant rule is *High ∧ Surplus ∧ Quiet ∧ Safe ∧ Hidden* → *Passive* (score 2.2). Despite the player being objectively strong, **Exposure's dominance holds Threat in the Passive band**, so the director sends almost nothing. The moment the player breaks cover and opens fire, Noise climbs and line-of-sight ramps Exposure toward *Spotted*; high-threat rules begin to fire and Threat rises smoothly toward Overwhelming — the swarm converges.

3.7 Fuzzy Modelling — Mamdani Inference, Step by Step

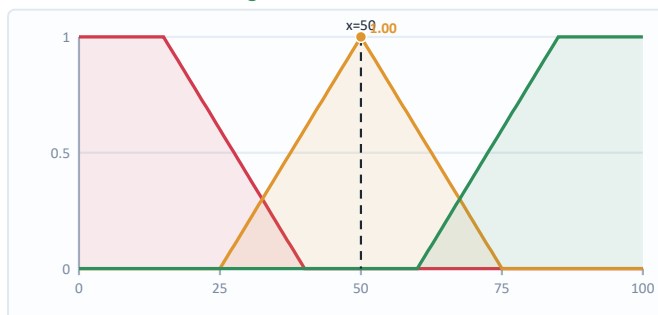
This section follows the **four-step Mamdani process** exactly as taught in the course (*Fuzzification* → *Rule Evaluation* → *Aggregation* → *Defuzzification*), applied to one concrete game state using the project's real membership functions and rule scoring. Sample state: **Health = 50, Ammo = 55, Noise = 55, Pressure = 15, Exposure = 70**.

STEP 1 — Fuzzification (crisp → fuzzy)

For each crisp input we (i) **take the crisp value**, (ii) **find its intersection** with the fuzzy-set curves, and (iii) **read off the degree of membership μ** on each term. The dashed marker is the crisp value; the coloured dots are the resulting μ :

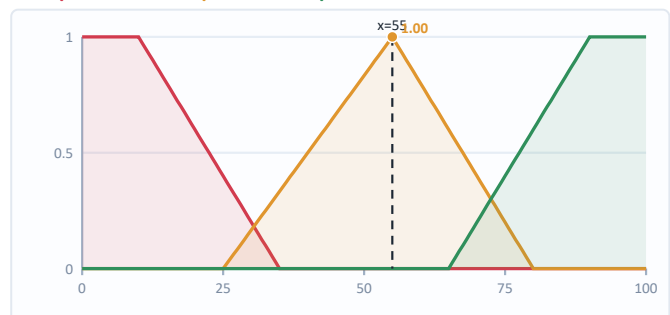
Player Health = 50

■ Low ■ Medium ■ High



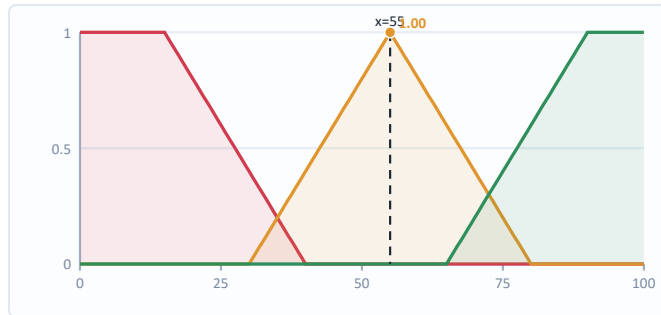
Ammo Count = 55

■ Depleted ■ Adequate ■ Surplus



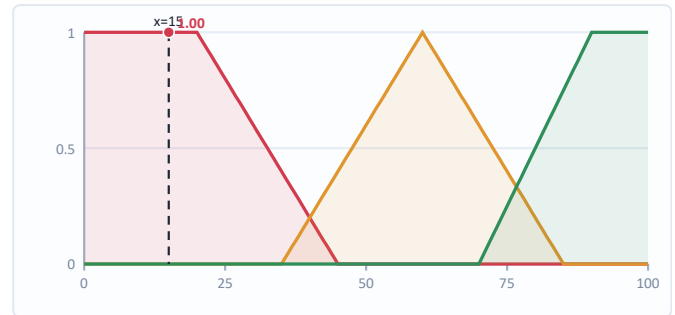
Noise Level = 55

■ Quiet ■ Moderate ■ Loud



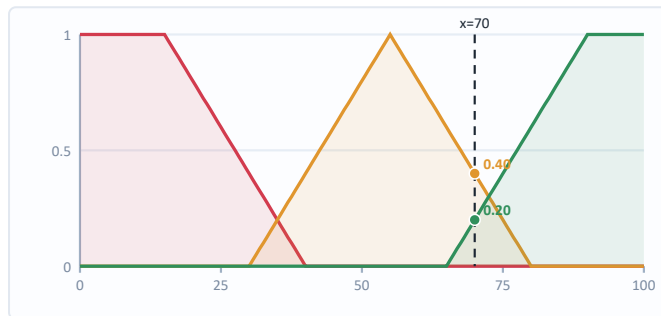
Pressure = 15

■ Safe ■ Engaged ■ Swarmed



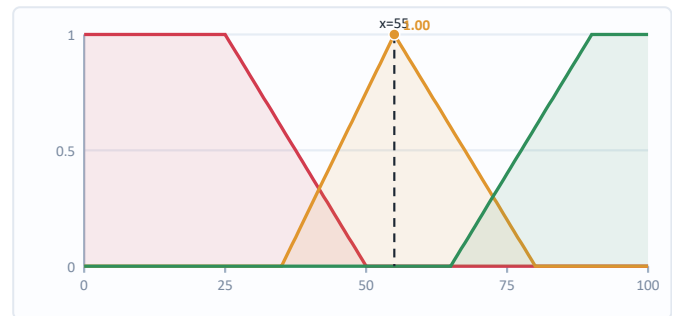
Exposure = 70

■ Hidden ■ Suspicious ■ Spotted



Player Skill = 55

■ Novice ■ Average ■ Expert



At this state every input except Exposure sits cleanly on a single term ($\mu = 1.0$ — including **Skill = Average 1.0**); only **Exposure straddles two terms** — Suspicious 0.40 and Spotted 0.20. Because each antecedent is combined with **AND = min** and the other five memberships are 1.0, the firing strength of every rule is decided by its Exposure term — so exactly **two of the 729 rules fire** (evaluated next in Step 2).

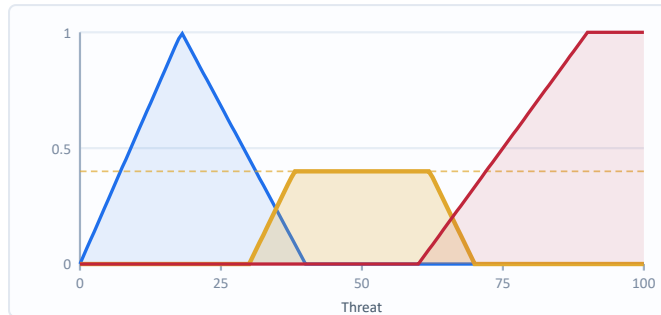
STEP 2 — Rule Evaluation (apply the fuzzy operators, then clip)

Each rule's antecedents are combined into one truth value: **AND uses min** ($\mu_{A \cap B} = \min[\mu_A, \mu_B]$) and **OR uses max** ($\mu_{A \cup B} = \max[\mu_A, \mu_B]$). That truth value ($\times$ the rule weight) then **clips** the consequent membership function — the output set is sliced flat at the firing level. Here all five non-Exposure memberships are 1.0, so each rule's strength is set by its Exposure term:

Rule	Antecedent — min of the six memberships	$\times$ weight w	= strength	THEN (clip)
R1	$\min(\text{Med } 1.0, \text{Adeq } 1.0, \text{Mod } 1.0, \text{Safe } 1.0, \text{Avg } 1.0, \text{Suspicious } 0.40) = 0.40$	$\times 1.00$	0.40	Tactical clipped at 0.40
R2	$\min(\text{Med } 1.0, \text{Adeq } 1.0, \text{Mod } 1.0, \text{Safe } 1.0, \text{Avg } 1.0, \text{Spotted } 0.20) = 0.20$	$\times 0.88$	0.176	Overwhelming clipped at 0.176

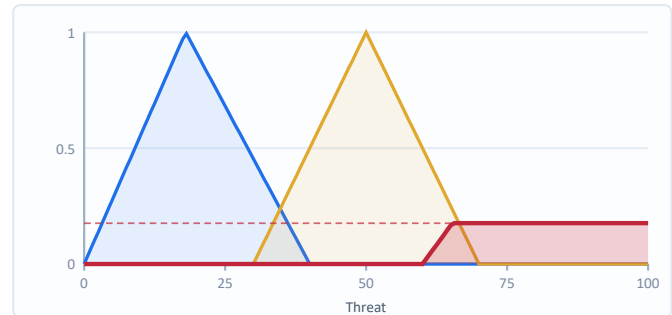
R1 → Tactical, clipped at 0.40

■ Passive ■ Tactical ■ Overwhelming



R2 → Overwhelming, clipped at 0.176

■ Passive ■ Tactical ■ Overwhelming

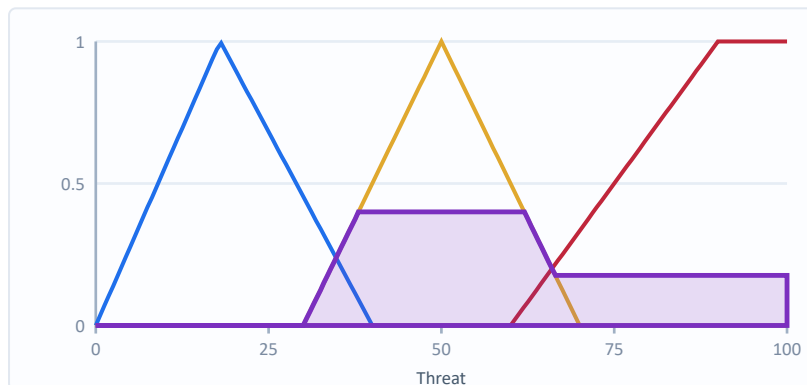


STEP 3 — Aggregation (unify the clipped consequents with OR = max)

The clipped output sets of all firing rules are unified into a single fuzzy set by taking the **maximum** at every point of the output universe — one aggregated Threat set:

Aggregated Threat set — OR = max of the clipped consequents

■ aggregated (union)



STEP 4 — Defuzzification (Centroid / Centre-of-Gravity)

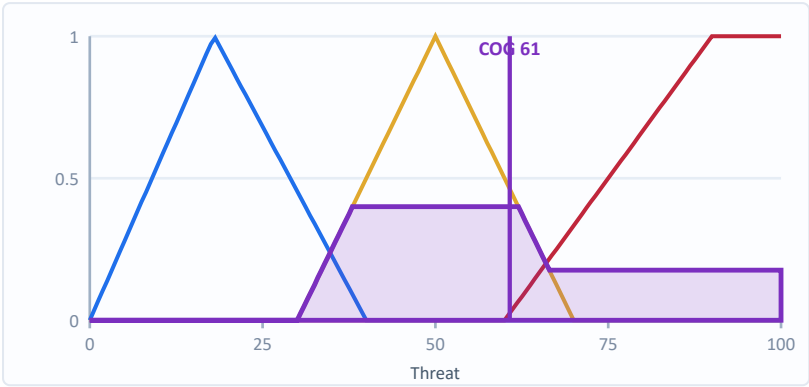
The most popular method is the **centroid (COG)** — the point where a vertical line splits the aggregated area into two equal masses. Estimated over a sample of points:

$$\text{COG} = \frac{\sum \mu(x) \cdot x}{\sum \mu(x)} = \frac{\sum \text{ over } x = 0, 2, \dots, 100 \text{ of } \mu(x) \cdot x}{\sum \text{ over } x \text{ of } \mu(x)} = 60.75 \rightarrow \text{Threat} \approx 61$$

So when the player is at this state, the crisp Threat ≈ **61** — the top of the Tactical band just entering Overwhelming. The director then turns this into a sizeable, fast wave (per §3.3).

Centroid (centre of gravity) of the aggregated area → crisp Threat

■ aggregated set | COG line



STEP 5 — The SECOND output: SUPPLY (same rules, second consequent)

The 2024 build added a **second Mamdani output, Supply**, defuzzified from the *same* firing rules in the *same* pass — the engine simply reads each rule's second consequent. The two rules that fired above both carry the Supply consequent **Balanced** (a Medium-health, Adequate-ammo, not-Swarmed player needs neither charity nor starving), so:

Rule	Strength (from Step 2)	Threat consequent (clip)	Supply consequent (clip)
R1	0.40	Tactical @ 0.40	Balanced @ 0.40
R2	0.176	Overwhelming @ 0.176	Balanced @ 0.176

Aggregating the Supply consequents (OR = max) gives **Balanced = max(0.40, 0.176) = 0.40**; the centroid of that symmetric clipped triangle is **Supply ≈ 50** — "keep drops normal". Because Threat and Supply are read from different consequent columns, they move *independently*: the worked-example table below shows how a *desperate* player (low health, low ammo, swarmed) instead clips **Generous** and pushes Supply ≈ 84, while a comfortable player clips **Scarce** ≈ 19 — the rubber-band that is deliberately separate from how hard the horde hits.

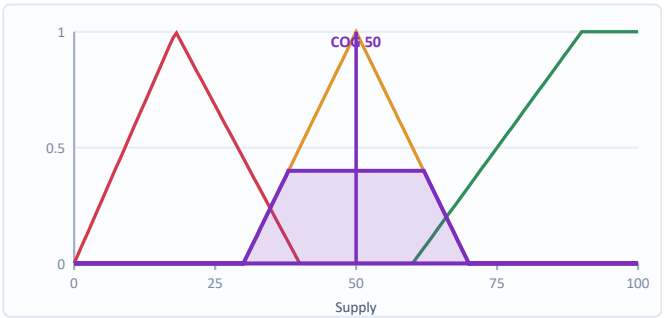
R1 & R2 → Supply Balanced, clipped at max(0.40, 0.176) = 0.40

■ Scarce ■ Balanced ■ Generous



Aggregate + Centroid → crisp Supply

■ Scarce ■ Balanced ■ Generous



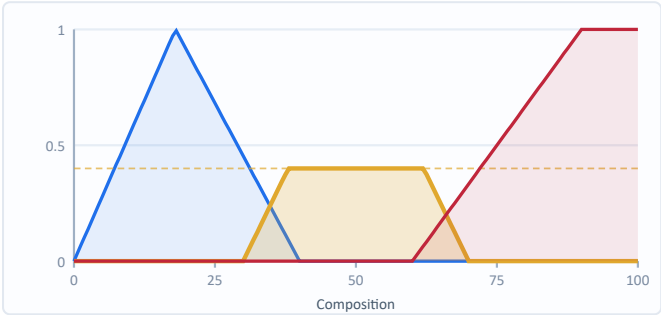
Player situation	Threat (output 1)	Supply (output 2)	Director's response
Desperate & hidden (HP 15, ammo 10, swarmed, unseen)	≈ 20 (Passive)	≈ 84 (Generous)	backs off the horde <i>and</i> showers pickups
Strong & spotted (HP 95, ammo 95, seen, loud)	≈ 86 (Overwhelming)	≈ 19 (Scarce)	full swarm, almost no pickups
This worked example (HP 50, ammo 55, ...)	≈ 61 (Tactical)	≈ 50 (Balanced)	steady pressure, normal drops

STEP 6 — The THIRD output: COMPOSITION (same rules, third consequent)

Phase 3 added a **third Mamdani output, Composition** — again read from the same firing rules in the same pass. It decides *what kind* of horde spawns (cheap grunts vs. brutes / spitters / elites), separately from how many. For this mid-game state both firing rules carry Composition **Mixed**, so it aggregates to **Mixed = 0.40** → centroid **Composition ≈ 50** ("a balanced enemy mix"). A skilled, spotted, loud player would instead clip **Heavy** (tougher elites), while a weak hidden player clips **Swarm** (fodder). The full worked example is therefore **Threat 61 · Supply 50 · Composition 50** — three independent decisions from one 729-rule pass:

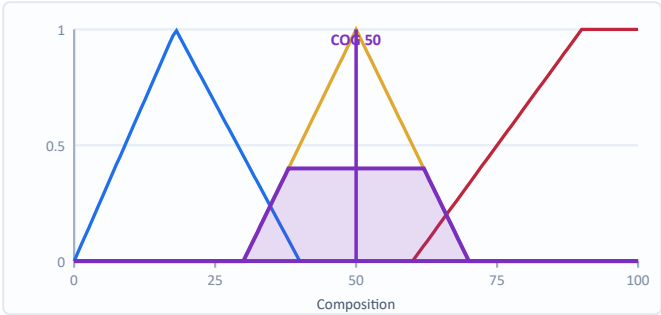
R1 & R2 → Composition Mixed, clipped at 0.40

■ Swarm ■ Mixed ■ Heavy



Aggregate + Centroid → crisp Composition

■ Swarm ■ Mixed ■ Heavy



The 6th input in action — SKILL adapts the difficulty

Skill enters the Threat policy as $+0.6 \cdot (\text{skill}-1)$, centred so **Average** is neutral. Holding the §3.7 state fixed and only varying Skill shows the adaptive difficulty directly (verified against the live engine):

Skill	Skill μ	Threat	Composition	Effect
Novice (15)	Novice 1.0	≈ 59	≈ 50	eased — the director backs off a struggling player
Average (55)	Average 1.0	≈ 61	≈ 50	baseline — identical to the pre-Phase-3 5-input build
Expert (95)	Expert 1.0	≈ 63.5	≈ 63.5	pushed — more Threat <i>and</i> a heavier enemy mix

3.8 Fuzzy Modelling — Takagi–Sugeno Inference, Step by Step

The same four-step process can be run in **Takagi–Sugeno (TSK)** style. As the lecture notes, Sugeno inference is *identical to Mamdani except for the rule consequent*: instead of a fuzzy set, each rule outputs a **crisp value** — a constant singleton **k** (zero-order) or a function $f(x,y)$ of the inputs (first-order). We run the very same §3.7 state (**Health 50, Ammo 55, Noise 55, Pressure 15, Exposure 70**).

STEP 1 — Fuzzification (identical to Mamdani)

Unchanged from §3.7: the crisp inputs give Med 1.0, Adeq 1.0, Mod 1.0, Safe 1.0, and Exposure Suspicious 0.40 / Spotted 0.20.

STEP 2 — Rule Evaluation (firing strength w , then a singleton spike — no clipping)

The antecedent truth value (firing strength w) is found exactly as before with AND = min × weight. But the consequent is now a **singleton spike** at a crisp output **k**, scaled to height w — there is no fuzzy set to clip. Map the three output bands to constant singletons (here, the centroids of the Mamdani sets): **Passive k = 19, Tactical k = 50, Overwhelming k = 86**.

Rule	Firing strength w (min × weight)	Singleton k	Spike
R1	0.40	$k = 50$ (Tactical)	spike of height 0.40 at $z = 50$
R2	0.176	$k = 86$ (Overwhelming)	spike of height 0.176 at $z = 86$

STEP 3 — Aggregation (collect the singleton spikes)

Aggregation simply gathers every rule's singleton spike on the output axis — no area to union. The graph below shows the two spikes (and the inactive Passive singleton).

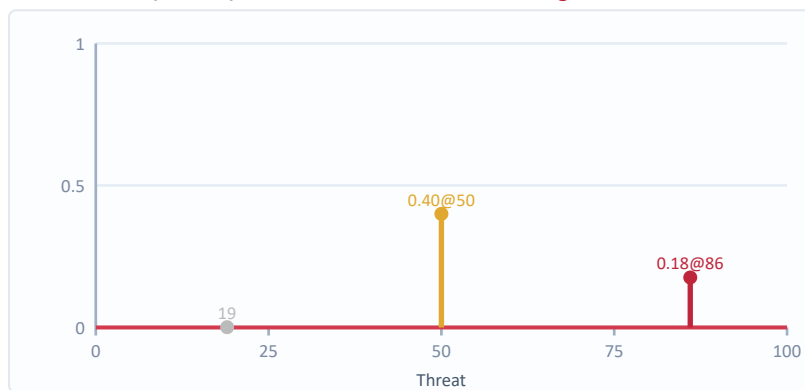
STEP 4 — Defuzzification (Weighted Average)

The crisp output is the **weighted average** of the singletons by their firing strengths (the standard Sugeno formula $WA = \frac{\sum \mu(k) \cdot k}{\sum \mu(k)}$):

$$WA = \frac{\mu(k_1) \cdot k_1 + \mu(k_2) \cdot k_2}{\mu(k_1) + \mu(k_2)} = \frac{0.40 \cdot 50 + 0.176 \cdot 86}{0.40 + 0.176} = \frac{35.14}{0.576} \approx 61.0$$

Sugeno — singleton spikes (height = firing strength)

■ Passive k=19 (inactive) ■ Tactical k=50 ■ Overwhelming k=86



The Sugeno result (≈ 61.0) sits within ~ 0.3 of the Mamdani centroid (**60.75**) — the two methods agree closely here, which is expected when the Sugeno singletons are taken as the Mamdani set centroids.

Aspect	Mamdani (<i>used in the game</i>)	Takagi–Sugeno (<i>this comparison</i>)
Fuzzification & AND	min of the 5 memberships $\times$ weight	identical
Consequent	a fuzzy set (Passive / Tactical / Overwhelming MF)	a crisp value z (constant, or a linear function of inputs)
Aggregation	OR = max (union of the clipped sets)	none — values go straight into a weighted average
Defuzzification	centroid of the aggregated area (sampled 0–100)	weighted average $\sum wz / \sum w$
Output (this state)	60.75	61.0
Strengths	intuitive, smooth, easy to <i>visualise</i> as a shape	cheap (no integration), easy to tune/optimize, great for control
Trade-offs	samples the output universe every tick	output is a number, not an interpretable fuzzy shape

Why we keep Mamdani: the live on-screen visualisation (clipped output sets + centroid) is central to the project's teaching goal, and Sugeno's singletons are harder to show as an intuitive shape. But because our rule base is already generated from a numeric *score*, switching to a **first-order Sugeno** (e.g. let each rule's consequent be $z = \text{remap}(\text{score} \rightarrow 0-100)$) would be a one-function change — a natural future option if raw speed or auto-tuning ever matters more than the visualisation.

3.9 The Game's Final Output — the Player's End Goal (Score · Kills · Wave)

It is important to separate two different kinds of "output":

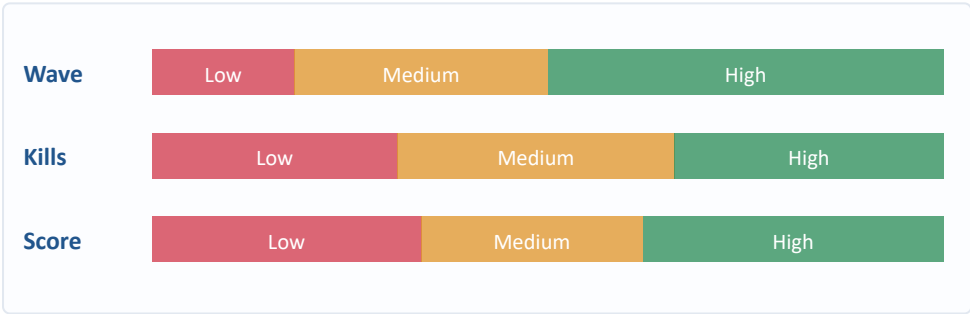
- **The fuzzy controller's output** is a single value — **Threat / Aggression** (§2.5, §3.7–3.8). Its job is to set the *difficulty*, not to score the player.
- **The game's final output** — what the player is actually trying to maximise — is the run result shown on the **OVERRUN (game-over) screen**: `Score · Kills · Wave` (drawn in `js/screens.js`). This is the endgame: *survive the fuzzy-scaled horde for as long as possible and rack up the highest Score / Kills / Wave*.

How the three are produced in `js/mechanics.js`:

Outcome metric	How it is earned
Kills	<code>G.kills++</code> on every enemy destroyed.
Score	<code>G.score += e.pts</code> — points by archetype: grunt 10 , runner 15 , brute 25 (tougher enemies are worth more).
Wave	<code>if (G.kills % 15 === 0) G.wave++</code> — the wave counter rises one step every 15 kills, so it tracks how deep the run went.

Because the fuzzy director keeps the challenge fair, a skilled player survives longer, which pushes all three metrics up together. To make a run easy to read at a glance, each metric can be banded **Low / Medium / High** (indicative targets for a default *Pace 0.6 / Density 1.0* run):

Run-result bands (reading aid — crisp scoreboard, not a fuzzy variable)



Band	Wave	Kills	Score	Interpretation
Low	1 – 3	0 – 30	0 – 500	early overrun — still learning the maze / noise discipline
Medium	4 – 7	31 – 100	501 – 1500	a solid run — managing health, ammo and exposure
High	8+	101+	1501+	mastery — using fog/stealth to control the horde

Note: Score / Kills / Wave are **crisp scoreboard metrics**, not fuzzy inputs or outputs of the controller — the Low/Medium/High labels above are a reading aid for run performance, kept deliberately separate from the six fuzzy inputs and three fuzzy outputs in §2.5.

3.10 Fuzzy Analytics — an In-Game Inspection Tool (Phase 3)

Pressing **C** in play opens a dedicated **Fuzzy Analytics** overlay that freezes the game and exposes the engine's internals — a teaching/verification instrument built directly on the live 729-rule controller:

- **Control surface** — a 2-D heat-map of Threat as a function of two chosen inputs (Exposure × Noise), the classic fuzzy control-surface plot, with a live marker at the current state.
- **Rule-firing heat-map** — all 729 rules as a 27×27 grid, each cell brightening with how often that rule has fired this run, so the whole base is inspectable at a glance.
- **Defuzzifier comparison** — the same aggregated Threat set defuzzified four ways side by side: **centroid**, **bisector**, **mean-of-maxima** and **Sugeno weighted average** — with a runtime **Mamdani ↔ Sugeno** toggle that switches

which method actually drives the game.

- **"Why this threat?" explainability** — one plain-English sentence generated from the dominant firing rule (also shown live under the rules panel).
- **Fuzzy weapon advisor** — a small separate FIS (nearest-enemy distance × crowd size → weapon) that hints which weapon suits the moment; fuzzy decision-support, not just difficulty.
- **CSV data-logger** — every inference logs the 6 inputs + 3 outputs; one click exports the whole run as a CSV for offline analysis / charts.

4. Requirements Compliance Checklist

Requirement	Where addressed	Status
Basic UI layout / interaction flow	§1.4 — setup, armory, HUD, minimap, gameplay, game-over	Met
Core functional components	§1.5 and §3.1	Met
Clear problem statement	§1.2 — stealth-aware dynamic difficulty balancing	Met
Target users	§1.3	Met
Rationale for fuzzy logic	§1.6 and §3	Met
Proposed fuzzy input variables	§2.1 — Health, Ammo, Noise, Pressure, Exposure, Skill (+ 3 outputs)	Exceeded (6 inputs)
Crisp → linguistic mapping (fuzzy graph)	§2.2 graphs (9 variables) + §3.7 full modelling diagrams	Met
Membership function ideas	§2.3 and §3.2	Met
At least 2 draft fuzzy rules	§2.4 + Appendix B — all 729 rules ; justified in §2.5 (combined rules, 3 ⁶)	Exceeded
Inference modelling (Mamdani & Sugeno)	§3.7 four-step Mamdani (×3 outputs) · §3.8 Takagi–Sugeno · §3.10 defuzzifier comparison	Exceeded
Input/output accounting & game outcome	§2.5 (6-in / 3-out MIMO) · §3.9 (Score / Kills / Wave)	Met

5. Self-Assessment vs Evaluation Criteria (15%)

Criterion	Weight	How this submission addresses it
Clarity & feasibility of idea	5%	A focused, well-scoped concept with a clear problem, named users and an already-running modular build — demonstrably feasible.
Prototype completeness (UI + functionality)	5%	A fully playable game (four development phases) with a working 729-rule fuzzy engine, procedural maze + stealth, perception AI, roguelite progression, full UX, audio, and an in-game fuzzy-analytics tool — well beyond a UI mock-up.
Fuzzy input identification & justification	5%	Six justified inputs and three outputs with universes, membership shapes, mapping graphs, a complete 729-rule generated base (Appendix B) with per-variable and rule-count justification (§2.5), and the pipeline shown Mamdani (§3.7), Sugeno (§3.8) and with a live defuzzifier comparison (§3.10).

6. How to Run the Prototype

- Open `index.html` in any modern browser (Chrome, Edge, Firefox) — plain script tags, no server needed (double-click works; or right-click → *Open with Live Server* in VS Code).

- On the **Mission Setup** screen, set the sliders (and optionally God mode), visit the **ARMORY** to spend coins, then click **DEPLOY**.
- Play with **WASD + mouse**; **Q / 1 / 2 / 3** switch weapons, **R** reloads, **G** toggles god mode, **B / V** toggle the rules/graphs panels, **Esc** pauses.
- Watch the **Threat gauge**, the **live membership-function graphs**, the **Active Fuzzy Rules** panel and the **minimap** respond live to your health, ammo, noise, how swarmed you are and how exposed you are. Hide in fog and stop firing to watch the horde lose interest.

Appendix A — Code Map: every file and what it does

The game is 17 JavaScript modules loaded in dependency order by `game/index.html` (plus CSS, the audio folder, and the report tooling). Each does one job:

File	Category	What it does
<code>index.html</code>	Loop/UI	The page: canvas, web-font, script load order, on-screen control hint.
<code>css/style.css</code>	UI	Page chrome only — the game UI is all canvas-drawn.
<code>js/config.js</code>	Loop	Canvas, <code>ctx</code> , logical <code>W / H</code> , responsive resize.
<code>js/balance.js</code>	Rendering/config	Central tuning constants (enemy cap, combo, dash, elite chance...) — edit here to rebalance.
<code>js/audio.js</code>	Audio	Procedural WebAudio SFX, streamed Molchat-Doma playlist, threat-dynamic music, beat detection.
<code>js/world.js</code>	World	Maze generation (seeded for the daily), collision, line-of-sight, fog, destructible walls, minimap prerender.
<code>js/fuzzy.js</code>	Fuzzy AI	The engine — 6 inputs, 3 outputs, 729-rule Mamdani + Sugeno, centroid/bisector/MoM defuzzifiers, rule heatmap, control-surface sampler, explainability; plus <code>MicroFuzzy</code> (per-enemy) and <code>WeaponAdvisor</code> .
<code>js/state.js</code>	State	The single <code>G</code> game-state object and <code>reset()</code> .
<code>js/weapons.js</code>	Progression	Weapon defs, levelling, power-ups, and the persistent meta (coins, unlocks, records, settings, achievements) in localStorage.
<code>js/roguelite.js</code>	Progression	Perks, sector modifiers, deployables (mine/turret), barrels, extraction beacon, achievements, per-sector setup.
<code>js/mechanics.js</code>	Simulation	The <code>update()</code> loop: input, movement, shooting, dash/melee, enemy AI + spawn director, the fuzzy-inference call, skill estimate, CSV log.
<code>js/input.js</code>	Input	Keyboard, mouse, gamepad polling, and touch virtual dual-stick.
<code>js/render.js</code>	Rendering	World, 2.5D walls + player-cast shadows, floor cache, lights, decals, dust, weather, gibs, characters, UI palette.
<code>js/post.js</code>	Rendering	Full-screen bloom post-process.
<code>js/hud.js</code>	HUD	Threat gauge, membership-function graphs, live rules + explainability, minimap, combo, deployables, boss bar, toasts.
<code>js/screens.js</code>	UI	Setup / armory / pause / death / settings / perk / tutorial screens and all click routing.
<code>js/analytics.js</code>	Fuzzy AI	The Fuzzy Analytics overlay (C): control surface, rule heatmap, defuzzifier comparison, CSV export, Mamdani↔Sugeno toggle.
<code>js/main.js</code>	Loop	The <code>requestAnimationFrame</code> driver: update → draw-by-state → overlays.

File	Category	What it does
<code>js/render3d.js</code> + <code>js/lib/three.min.js</code>	—	<i>Experimental</i> first-person three.js renderer (not loaded by default).

Report tooling (in `report/`): `_report_src.html` is this document's template; `report_build.js` generates all the SVG fuzzy graphs and the 729-rule Appendix B and injects them, producing `Progress_Report_1_UPDATED.html` (Print → Save as PDF).

Appendix B — Complete 729-Rule Base

Every rule below is generated by `js/fuzzy.js` from the three scoring policies in §2.4. Each of the $3^6 = 729$ antecedent combinations (Health × Ammo × Noise × Pressure × Exposure × Skill) is listed once, in engine build order, with all **three** defuzzified consequents (Threat, Supply, Composition) and the firing weight `w`. Every antecedent is joined by **AND (min)**.

Threat totals: 271 Passive · 308 Tactical · 150 Overwhelming. Supply totals: 189 Scarce · 270 Balanced · 270

Generous. Composition totals: 156 Swarm · 408 Mixed · 165 Heavy. (All three consequents share the same 729 antecedents — three policies, one grid.)

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
1	Low	Depleted	Quiet	Safe	Hidden	Novice	-0.60	Passive	Generous	Swarm	1.00
2	Low	Depleted	Quiet	Safe	Hidden	Average	0.00	Passive	Generous	Swarm	1.00
3	Low	Depleted	Quiet	Safe	Hidden	Expert	0.60	Passive	Generous	Swarm	1.00
4	Low	Depleted	Quiet	Safe	Suspicious	Novice	1.10	Passive	Generous	Swarm	1.00
5	Low	Depleted	Quiet	Safe	Suspicious	Average	1.70	Passive	Generous	Swarm	0.84
6	Low	Depleted	Quiet	Safe	Suspicious	Expert	2.30	Passive	Generous	Swarm	0.60
7	Low	Depleted	Quiet	Safe	Spotted	Novice	2.80	Tactical	Generous	Swarm	0.80
8	Low	Depleted	Quiet	Safe	Spotted	Average	3.40	Tactical	Generous	Mixed	1.00
9	Low	Depleted	Quiet	Safe	Spotted	Expert	4.00	Tactical	Generous	Mixed	0.84
10	Low	Depleted	Quiet	Engaged	Hidden	Novice	-1.20	Passive	Generous	Swarm	1.00
11	Low	Depleted	Quiet	Engaged	Hidden	Average	-0.60	Passive	Generous	Swarm	1.00
12	Low	Depleted	Quiet	Engaged	Hidden	Expert	0.00	Passive	Generous	Swarm	1.00
13	Low	Depleted	Quiet	Engaged	Suspicious	Novice	0.50	Passive	Generous	Swarm	1.00
14	Low	Depleted	Quiet	Engaged	Suspicious	Average	1.10	Passive	Generous	Swarm	1.00
15	Low	Depleted	Quiet	Engaged	Suspicious	Expert	1.70	Passive	Generous	Swarm	0.84
16	Low	Depleted	Quiet	Engaged	Spotted	Novice	2.20	Passive	Generous	Swarm	0.64
17	Low	Depleted	Quiet	Engaged	Spotted	Average	2.80	Tactical	Generous	Mixed	0.80
18	Low	Depleted	Quiet	Engaged	Spotted	Expert	3.40	Tactical	Generous	Mixed	1.00
19	Low	Depleted	Quiet	Swarmed	Hidden	Novice	-1.80	Passive	Generous	Swarm	1.00
20	Low	Depleted	Quiet	Swarmed	Hidden	Average	-1.20	Passive	Generous	Swarm	1.00
21	Low	Depleted	Quiet	Swarmed	Hidden	Expert	-0.60	Passive	Generous	Swarm	1.00
22	Low	Depleted	Quiet	Swarmed	Suspicious	Novice	-0.10	Passive	Generous	Swarm	1.00
23	Low	Depleted	Quiet	Swarmed	Suspicious	Average	0.50	Passive	Generous	Swarm	1.00
24	Low	Depleted	Quiet	Swarmed	Suspicious	Expert	1.10	Passive	Generous	Swarm	1.00
25	Low	Depleted	Quiet	Swarmed	Spotted	Novice	1.60	Passive	Generous	Swarm	0.88
26	Low	Depleted	Quiet	Swarmed	Spotted	Average	2.20	Passive	Generous	Mixed	0.64
27	Low	Depleted	Quiet	Swarmed	Spotted	Expert	2.80	Tactical	Generous	Mixed	0.80
28	Low	Depleted	Moderate	Safe	Hidden	Novice	0.20	Passive	Generous	Swarm	1.00
29	Low	Depleted	Moderate	Safe	Hidden	Average	0.80	Passive	Generous	Swarm	1.00
30	Low	Depleted	Moderate	Safe	Hidden	Expert	1.40	Passive	Generous	Swarm	0.96
31	Low	Depleted	Moderate	Safe	Suspicious	Novice	1.90	Passive	Generous	Swarm	0.76
32	Low	Depleted	Moderate	Safe	Suspicious	Average	2.50	Tactical	Generous	Swarm	0.68
33	Low	Depleted	Moderate	Safe	Suspicious	Expert	3.10	Tactical	Generous	Mixed	0.92
34	Low	Depleted	Moderate	Safe	Spotted	Novice	3.60	Tactical	Generous	Mixed	1.00
35	Low	Depleted	Moderate	Safe	Spotted	Average	4.20	Tactical	Generous	Mixed	0.76
36	Low	Depleted	Moderate	Safe	Spotted	Expert	4.80	Overwhelming	Generous	Mixed	0.68
37	Low	Depleted	Moderate	Engaged	Hidden	Novice	-0.40	Passive	Generous	Swarm	1.00

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
38	Low	Depleted	Moderate	Engaged	Hidden	Average	0.20	Passive	Generous	Swarm	1.00
39	Low	Depleted	Moderate	Engaged	Hidden	Expert	0.80	Passive	Generous	Swarm	1.00
40	Low	Depleted	Moderate	Engaged	Suspicious	Novice	1.30	Passive	Generous	Swarm	1.00
41	Low	Depleted	Moderate	Engaged	Suspicious	Average	1.90	Passive	Generous	Swarm	0.76
42	Low	Depleted	Moderate	Engaged	Suspicious	Expert	2.50	Tactical	Generous	Mixed	0.68
43	Low	Depleted	Moderate	Engaged	Spotted	Novice	3.00	Tactical	Generous	Mixed	0.88
44	Low	Depleted	Moderate	Engaged	Spotted	Average	3.60	Tactical	Generous	Mixed	1.00
45	Low	Depleted	Moderate	Engaged	Spotted	Expert	4.20	Tactical	Generous	Mixed	0.76
46	Low	Depleted	Moderate	Swarmed	Hidden	Novice	-1.00	Passive	Generous	Swarm	1.00
47	Low	Depleted	Moderate	Swarmed	Hidden	Average	-0.40	Passive	Generous	Swarm	1.00
48	Low	Depleted	Moderate	Swarmed	Hidden	Expert	0.20	Passive	Generous	Swarm	1.00
49	Low	Depleted	Moderate	Swarmed	Suspicious	Novice	0.70	Passive	Generous	Swarm	1.00
50	Low	Depleted	Moderate	Swarmed	Suspicious	Average	1.30	Passive	Generous	Swarm	1.00
51	Low	Depleted	Moderate	Swarmed	Suspicious	Expert	1.90	Passive	Generous	Mixed	0.76
52	Low	Depleted	Moderate	Swarmed	Spotted	Novice	2.40	Tactical	Generous	Mixed	0.64
53	Low	Depleted	Moderate	Swarmed	Spotted	Average	3.00	Tactical	Generous	Mixed	0.88
54	Low	Depleted	Moderate	Swarmed	Spotted	Expert	3.60	Tactical	Generous	Mixed	1.00
55	Low	Depleted	Loud	Safe	Hidden	Novice	1.00	Passive	Generous	Swarm	1.00
56	Low	Depleted	Loud	Safe	Hidden	Average	1.60	Passive	Generous	Swarm	0.88
57	Low	Depleted	Loud	Safe	Hidden	Expert	2.20	Passive	Generous	Mixed	0.64
58	Low	Depleted	Loud	Safe	Suspicious	Novice	2.70	Tactical	Generous	Mixed	0.76
59	Low	Depleted	Loud	Safe	Suspicious	Average	3.30	Tactical	Generous	Mixed	1.00
60	Low	Depleted	Loud	Safe	Suspicious	Expert	3.90	Tactical	Generous	Mixed	0.88
61	Low	Depleted	Loud	Safe	Spotted	Novice	4.40	Tactical	Generous	Mixed	0.68
62	Low	Depleted	Loud	Safe	Spotted	Average	5.00	Overwhelming	Generous	Mixed	0.76
63	Low	Depleted	Loud	Safe	Spotted	Expert	5.60	Overwhelming	Generous	Heavy	1.00
64	Low	Depleted	Loud	Engaged	Hidden	Novice	0.40	Passive	Generous	Swarm	1.00
65	Low	Depleted	Loud	Engaged	Hidden	Average	1.00	Passive	Generous	Swarm	1.00
66	Low	Depleted	Loud	Engaged	Hidden	Expert	1.60	Passive	Generous	Mixed	0.88
67	Low	Depleted	Loud	Engaged	Suspicious	Novice	2.10	Passive	Generous	Mixed	0.68
68	Low	Depleted	Loud	Engaged	Suspicious	Average	2.70	Tactical	Generous	Mixed	0.76
69	Low	Depleted	Loud	Engaged	Suspicious	Expert	3.30	Tactical	Generous	Mixed	1.00
70	Low	Depleted	Loud	Engaged	Spotted	Novice	3.80	Tactical	Generous	Mixed	0.92
71	Low	Depleted	Loud	Engaged	Spotted	Average	4.40	Tactical	Generous	Mixed	0.68
72	Low	Depleted	Loud	Engaged	Spotted	Expert	5.00	Overwhelming	Generous	Heavy	0.76
73	Low	Depleted	Loud	Swarmed	Hidden	Novice	-0.20	Passive	Generous	Swarm	1.00
74	Low	Depleted	Loud	Swarmed	Hidden	Average	0.40	Passive	Generous	Swarm	1.00
75	Low	Depleted	Loud	Swarmed	Hidden	Expert	1.00	Passive	Generous	Mixed	1.00
76	Low	Depleted	Loud	Swarmed	Suspicious	Novice	1.50	Passive	Generous	Mixed	0.92
77	Low	Depleted	Loud	Swarmed	Suspicious	Average	2.10	Passive	Generous	Mixed	0.68
78	Low	Depleted	Loud	Swarmed	Suspicious	Expert	2.70	Tactical	Generous	Mixed	0.76
79	Low	Depleted	Loud	Swarmed	Spotted	Novice	3.20	Tactical	Generous	Mixed	0.96
80	Low	Depleted	Loud	Swarmed	Spotted	Average	3.80	Tactical	Generous	Mixed	0.92
81	Low	Depleted	Loud	Swarmed	Spotted	Expert	4.40	Tactical	Generous	Heavy	0.68
82	Low	Adequate	Quiet	Safe	Hidden	Novice	-0.05	Passive	Balanced	Swarm	1.00
83	Low	Adequate	Quiet	Safe	Hidden	Average	0.55	Passive	Balanced	Swarm	1.00
84	Low	Adequate	Quiet	Safe	Hidden	Expert	1.15	Passive	Balanced	Swarm	1.00

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
85	Low	Adequate	Quiet	Safe	Suspicious	Novice	1.65	Passive	Balanced	Swarm	0.86
86	Low	Adequate	Quiet	Safe	Suspicious	Average	2.25	Passive	Balanced	Swarm	0.62
87	Low	Adequate	Quiet	Safe	Suspicious	Expert	2.85	Tactical	Balanced	Mixed	0.82
88	Low	Adequate	Quiet	Safe	Spotted	Novice	3.35	Tactical	Balanced	Mixed	1.00
89	Low	Adequate	Quiet	Safe	Spotted	Average	3.95	Tactical	Balanced	Mixed	0.86
90	Low	Adequate	Quiet	Safe	Spotted	Expert	4.55	Tactical	Balanced	Mixed	0.62
91	Low	Adequate	Quiet	Engaged	Hidden	Novice	-0.65	Passive	Generous	Swarm	1.00
92	Low	Adequate	Quiet	Engaged	Hidden	Average	-0.05	Passive	Generous	Swarm	1.00
93	Low	Adequate	Quiet	Engaged	Hidden	Expert	0.55	Passive	Generous	Swarm	1.00
94	Low	Adequate	Quiet	Engaged	Suspicious	Novice	1.05	Passive	Generous	Swarm	1.00
95	Low	Adequate	Quiet	Engaged	Suspicious	Average	1.65	Passive	Generous	Swarm	0.86
96	Low	Adequate	Quiet	Engaged	Suspicious	Expert	2.25	Passive	Generous	Mixed	0.62
97	Low	Adequate	Quiet	Engaged	Spotted	Novice	2.75	Tactical	Generous	Mixed	0.78
98	Low	Adequate	Quiet	Engaged	Spotted	Average	3.35	Tactical	Generous	Mixed	1.00
99	Low	Adequate	Quiet	Engaged	Spotted	Expert	3.95	Tactical	Generous	Mixed	0.86
100	Low	Adequate	Quiet	Swarmed	Hidden	Novice	-1.25	Passive	Generous	Swarm	1.00
101	Low	Adequate	Quiet	Swarmed	Hidden	Average	-0.65	Passive	Generous	Swarm	1.00
102	Low	Adequate	Quiet	Swarmed	Hidden	Expert	-0.05	Passive	Generous	Swarm	1.00
103	Low	Adequate	Quiet	Swarmed	Suspicious	Novice	0.45	Passive	Generous	Swarm	1.00
104	Low	Adequate	Quiet	Swarmed	Suspicious	Average	1.05	Passive	Generous	Swarm	1.00
105	Low	Adequate	Quiet	Swarmed	Suspicious	Expert	1.65	Passive	Generous	Mixed	0.86
106	Low	Adequate	Quiet	Swarmed	Spotted	Novice	2.15	Passive	Generous	Mixed	0.66
107	Low	Adequate	Quiet	Swarmed	Spotted	Average	2.75	Tactical	Generous	Mixed	0.78
108	Low	Adequate	Quiet	Swarmed	Spotted	Expert	3.35	Tactical	Generous	Mixed	1.00
109	Low	Adequate	Moderate	Safe	Hidden	Novice	0.75	Passive	Balanced	Swarm	1.00
110	Low	Adequate	Moderate	Safe	Hidden	Average	1.35	Passive	Balanced	Swarm	0.98
111	Low	Adequate	Moderate	Safe	Hidden	Expert	1.95	Passive	Balanced	Mixed	0.74
112	Low	Adequate	Moderate	Safe	Suspicious	Novice	2.45	Tactical	Balanced	Swarm	0.66
113	Low	Adequate	Moderate	Safe	Suspicious	Average	3.05	Tactical	Balanced	Mixed	0.90
114	Low	Adequate	Moderate	Safe	Suspicious	Expert	3.65	Tactical	Balanced	Mixed	0.98
115	Low	Adequate	Moderate	Safe	Spotted	Novice	4.15	Tactical	Balanced	Mixed	0.78
116	Low	Adequate	Moderate	Safe	Spotted	Average	4.75	Overwhelming	Balanced	Mixed	0.66
117	Low	Adequate	Moderate	Safe	Spotted	Expert	5.35	Overwhelming	Balanced	Mixed	0.90
118	Low	Adequate	Moderate	Engaged	Hidden	Novice	0.15	Passive	Generous	Swarm	1.00
119	Low	Adequate	Moderate	Engaged	Hidden	Average	0.75	Passive	Generous	Swarm	1.00
120	Low	Adequate	Moderate	Engaged	Hidden	Expert	1.35	Passive	Generous	Mixed	0.98
121	Low	Adequate	Moderate	Engaged	Suspicious	Novice	1.85	Passive	Generous	Swarm	0.78
122	Low	Adequate	Moderate	Engaged	Suspicious	Average	2.45	Tactical	Generous	Mixed	0.66
123	Low	Adequate	Moderate	Engaged	Suspicious	Expert	3.05	Tactical	Generous	Mixed	0.90
124	Low	Adequate	Moderate	Engaged	Spotted	Novice	3.55	Tactical	Generous	Mixed	1.00
125	Low	Adequate	Moderate	Engaged	Spotted	Average	4.15	Tactical	Generous	Mixed	0.78
126	Low	Adequate	Moderate	Engaged	Spotted	Expert	4.75	Overwhelming	Generous	Mixed	0.66
127	Low	Adequate	Moderate	Swarmed	Hidden	Novice	-0.45	Passive	Generous	Swarm	1.00
128	Low	Adequate	Moderate	Swarmed	Hidden	Average	0.15	Passive	Generous	Swarm	1.00
129	Low	Adequate	Moderate	Swarmed	Hidden	Expert	0.75	Passive	Generous	Mixed	1.00
130	Low	Adequate	Moderate	Swarmed	Suspicious	Novice	1.25	Passive	Generous	Swarm	1.00
131	Low	Adequate	Moderate	Swarmed	Suspicious	Average	1.85	Passive	Generous	Mixed	0.78

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
132	Low	Adequate	Moderate	Swarmed	Suspicious	Expert	2.45	Tactical	Generous	Mixed	0.66
133	Low	Adequate	Moderate	Swarmed	Spotted	Novice	2.95	Tactical	Generous	Mixed	0.86
134	Low	Adequate	Moderate	Swarmed	Spotted	Average	3.55	Tactical	Generous	Mixed	1.00
135	Low	Adequate	Moderate	Swarmed	Spotted	Expert	4.15	Tactical	Generous	Mixed	0.78
136	Low	Adequate	Loud	Safe	Hidden	Novice	1.55	Passive	Balanced	Swarm	0.90
137	Low	Adequate	Loud	Safe	Hidden	Average	2.15	Passive	Balanced	Mixed	0.66
138	Low	Adequate	Loud	Safe	Hidden	Expert	2.75	Tactical	Balanced	Mixed	0.78
139	Low	Adequate	Loud	Safe	Suspicious	Novice	3.25	Tactical	Balanced	Mixed	0.98
140	Low	Adequate	Loud	Safe	Suspicious	Average	3.85	Tactical	Balanced	Mixed	0.90
141	Low	Adequate	Loud	Safe	Suspicious	Expert	4.45	Tactical	Balanced	Mixed	0.66
142	Low	Adequate	Loud	Safe	Spotted	Novice	4.95	Overwhelming	Balanced	Mixed	0.74
143	Low	Adequate	Loud	Safe	Spotted	Average	5.55	Overwhelming	Balanced	Heavy	0.98
144	Low	Adequate	Loud	Safe	Spotted	Expert	6.15	Overwhelming	Balanced	Heavy	1.00
145	Low	Adequate	Loud	Engaged	Hidden	Novice	0.95	Passive	Generous	Swarm	1.00
146	Low	Adequate	Loud	Engaged	Hidden	Average	1.55	Passive	Generous	Mixed	0.90
147	Low	Adequate	Loud	Engaged	Hidden	Expert	2.15	Passive	Generous	Mixed	0.66
148	Low	Adequate	Loud	Engaged	Suspicious	Novice	2.65	Tactical	Generous	Mixed	0.74
149	Low	Adequate	Loud	Engaged	Suspicious	Average	3.25	Tactical	Generous	Mixed	0.98
150	Low	Adequate	Loud	Engaged	Suspicious	Expert	3.85	Tactical	Generous	Mixed	0.90
151	Low	Adequate	Loud	Engaged	Spotted	Novice	4.35	Tactical	Generous	Mixed	0.70
152	Low	Adequate	Loud	Engaged	Spotted	Average	4.95	Overwhelming	Generous	Heavy	0.74
153	Low	Adequate	Loud	Engaged	Spotted	Expert	5.55	Overwhelming	Generous	Heavy	0.98
154	Low	Adequate	Loud	Swarmed	Hidden	Novice	0.35	Passive	Generous	Swarm	1.00
155	Low	Adequate	Loud	Swarmed	Hidden	Average	0.95	Passive	Generous	Mixed	1.00
156	Low	Adequate	Loud	Swarmed	Hidden	Expert	1.55	Passive	Generous	Mixed	0.90
157	Low	Adequate	Loud	Swarmed	Suspicious	Novice	2.05	Passive	Generous	Mixed	0.70
158	Low	Adequate	Loud	Swarmed	Suspicious	Average	2.65	Tactical	Generous	Mixed	0.74
159	Low	Adequate	Loud	Swarmed	Suspicious	Expert	3.25	Tactical	Generous	Mixed	0.98
160	Low	Adequate	Loud	Swarmed	Spotted	Novice	3.75	Tactical	Generous	Mixed	0.94
161	Low	Adequate	Loud	Swarmed	Spotted	Average	4.35	Tactical	Generous	Heavy	0.70
162	Low	Adequate	Loud	Swarmed	Spotted	Expert	4.95	Overwhelming	Generous	Heavy	0.74
163	Low	Surplus	Quiet	Safe	Hidden	Novice	0.50	Passive	Balanced	Swarm	1.00
164	Low	Surplus	Quiet	Safe	Hidden	Average	1.10	Passive	Balanced	Swarm	1.00
165	Low	Surplus	Quiet	Safe	Hidden	Expert	1.70	Passive	Balanced	Swarm	0.84
166	Low	Surplus	Quiet	Safe	Suspicious	Novice	2.20	Passive	Balanced	Swarm	0.64
167	Low	Surplus	Quiet	Safe	Suspicious	Average	2.80	Tactical	Balanced	Mixed	0.80
168	Low	Surplus	Quiet	Safe	Suspicious	Expert	3.40	Tactical	Balanced	Mixed	1.00
169	Low	Surplus	Quiet	Safe	Spotted	Novice	3.90	Tactical	Balanced	Mixed	0.88
170	Low	Surplus	Quiet	Safe	Spotted	Average	4.50	Tactical	Balanced	Mixed	0.64
171	Low	Surplus	Quiet	Safe	Spotted	Expert	5.10	Overwhelming	Balanced	Mixed	0.80
172	Low	Surplus	Quiet	Engaged	Hidden	Novice	-0.10	Passive	Balanced	Swarm	1.00
173	Low	Surplus	Quiet	Engaged	Hidden	Average	0.50	Passive	Balanced	Swarm	1.00
174	Low	Surplus	Quiet	Engaged	Hidden	Expert	1.10	Passive	Balanced	Swarm	1.00
175	Low	Surplus	Quiet	Engaged	Suspicious	Novice	1.60	Passive	Balanced	Swarm	0.88
176	Low	Surplus	Quiet	Engaged	Suspicious	Average	2.20	Passive	Balanced	Mixed	0.64
177	Low	Surplus	Quiet	Engaged	Suspicious	Expert	2.80	Tactical	Balanced	Mixed	0.80
178	Low	Surplus	Quiet	Engaged	Spotted	Novice	3.30	Tactical	Balanced	Mixed	1.00

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
179	Low	Surplus	Quiet	Engaged	Spotted	Average	3.90	Tactical	Balanced	Mixed	0.88
180	Low	Surplus	Quiet	Engaged	Spotted	Expert	4.50	Tactical	Balanced	Mixed	0.64
181	Low	Surplus	Quiet	Swarmed	Hidden	Novice	-0.70	Passive	Generous	Swarm	1.00
182	Low	Surplus	Quiet	Swarmed	Hidden	Average	-0.10	Passive	Generous	Swarm	1.00
183	Low	Surplus	Quiet	Swarmed	Hidden	Expert	0.50	Passive	Generous	Swarm	1.00
184	Low	Surplus	Quiet	Swarmed	Suspicious	Novice	1.00	Passive	Generous	Swarm	1.00
185	Low	Surplus	Quiet	Swarmed	Suspicious	Average	1.60	Passive	Generous	Mixed	0.88
186	Low	Surplus	Quiet	Swarmed	Suspicious	Expert	2.20	Passive	Generous	Mixed	0.64
187	Low	Surplus	Quiet	Swarmed	Spotted	Novice	2.70	Tactical	Generous	Mixed	0.76
188	Low	Surplus	Quiet	Swarmed	Spotted	Average	3.30	Tactical	Generous	Mixed	1.00
189	Low	Surplus	Quiet	Swarmed	Spotted	Expert	3.90	Tactical	Generous	Mixed	0.88
190	Low	Surplus	Moderate	Safe	Hidden	Novice	1.30	Passive	Balanced	Swarm	1.00
191	Low	Surplus	Moderate	Safe	Hidden	Average	1.90	Passive	Balanced	Mixed	0.76
192	Low	Surplus	Moderate	Safe	Hidden	Expert	2.50	Tactical	Balanced	Mixed	0.68
193	Low	Surplus	Moderate	Safe	Suspicious	Novice	3.00	Tactical	Balanced	Mixed	0.88
194	Low	Surplus	Moderate	Safe	Suspicious	Average	3.60	Tactical	Balanced	Mixed	1.00
195	Low	Surplus	Moderate	Safe	Suspicious	Expert	4.20	Tactical	Balanced	Mixed	0.76
196	Low	Surplus	Moderate	Safe	Spotted	Novice	4.70	Overwhelming	Balanced	Mixed	0.64
197	Low	Surplus	Moderate	Safe	Spotted	Average	5.30	Overwhelming	Balanced	Mixed	0.88
198	Low	Surplus	Moderate	Safe	Spotted	Expert	5.90	Overwhelming	Balanced	Heavy	1.00
199	Low	Surplus	Moderate	Engaged	Hidden	Novice	0.70	Passive	Balanced	Swarm	1.00
200	Low	Surplus	Moderate	Engaged	Hidden	Average	1.30	Passive	Balanced	Mixed	1.00
201	Low	Surplus	Moderate	Engaged	Hidden	Expert	1.90	Passive	Balanced	Mixed	0.76
202	Low	Surplus	Moderate	Engaged	Suspicious	Novice	2.40	Tactical	Balanced	Mixed	0.64
203	Low	Surplus	Moderate	Engaged	Suspicious	Average	3.00	Tactical	Balanced	Mixed	0.88
204	Low	Surplus	Moderate	Engaged	Suspicious	Expert	3.60	Tactical	Balanced	Mixed	1.00
205	Low	Surplus	Moderate	Engaged	Spotted	Novice	4.10	Tactical	Balanced	Mixed	0.80
206	Low	Surplus	Moderate	Engaged	Spotted	Average	4.70	Overwhelming	Balanced	Mixed	0.64
207	Low	Surplus	Moderate	Engaged	Spotted	Expert	5.30	Overwhelming	Balanced	Heavy	0.88
208	Low	Surplus	Moderate	Swarmed	Hidden	Novice	0.10	Passive	Generous	Swarm	1.00
209	Low	Surplus	Moderate	Swarmed	Hidden	Average	0.70	Passive	Generous	Mixed	1.00
210	Low	Surplus	Moderate	Swarmed	Hidden	Expert	1.30	Passive	Generous	Mixed	1.00
211	Low	Surplus	Moderate	Swarmed	Suspicious	Novice	1.80	Passive	Generous	Mixed	0.80
212	Low	Surplus	Moderate	Swarmed	Suspicious	Average	2.40	Tactical	Generous	Mixed	0.64
213	Low	Surplus	Moderate	Swarmed	Suspicious	Expert	3.00	Tactical	Generous	Mixed	0.88
214	Low	Surplus	Moderate	Swarmed	Spotted	Novice	3.50	Tactical	Generous	Mixed	1.00
215	Low	Surplus	Moderate	Swarmed	Spotted	Average	4.10	Tactical	Generous	Mixed	0.80
216	Low	Surplus	Moderate	Swarmed	Spotted	Expert	4.70	Overwhelming	Generous	Heavy	0.64
217	Low	Surplus	Loud	Safe	Hidden	Novice	2.10	Passive	Balanced	Mixed	0.68
218	Low	Surplus	Loud	Safe	Hidden	Average	2.70	Tactical	Balanced	Mixed	0.76
219	Low	Surplus	Loud	Safe	Hidden	Expert	3.30	Tactical	Balanced	Mixed	1.00
220	Low	Surplus	Loud	Safe	Suspicious	Novice	3.80	Tactical	Balanced	Mixed	0.92
221	Low	Surplus	Loud	Safe	Suspicious	Average	4.40	Tactical	Balanced	Mixed	0.68
222	Low	Surplus	Loud	Safe	Suspicious	Expert	5.00	Overwhelming	Balanced	Heavy	0.76
223	Low	Surplus	Loud	Safe	Spotted	Novice	5.50	Overwhelming	Balanced	Heavy	0.96
224	Low	Surplus	Loud	Safe	Spotted	Average	6.10	Overwhelming	Balanced	Heavy	1.00
225	Low	Surplus	Loud	Safe	Spotted	Expert	6.70	Overwhelming	Balanced	Heavy	1.00

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
226	Low	Surplus	Loud	Engaged	Hidden	Novice	1.50	Passive	Balanced	Mixed	0.92
227	Low	Surplus	Loud	Engaged	Hidden	Average	2.10	Passive	Balanced	Mixed	0.68
228	Low	Surplus	Loud	Engaged	Hidden	Expert	2.70	Tactical	Balanced	Mixed	0.76
229	Low	Surplus	Loud	Engaged	Suspicious	Novice	3.20	Tactical	Balanced	Mixed	0.96
230	Low	Surplus	Loud	Engaged	Suspicious	Average	3.80	Tactical	Balanced	Mixed	0.92
231	Low	Surplus	Loud	Engaged	Suspicious	Expert	4.40	Tactical	Balanced	Heavy	0.68
232	Low	Surplus	Loud	Engaged	Spotted	Novice	4.90	Overwhelming	Balanced	Heavy	0.72
233	Low	Surplus	Loud	Engaged	Spotted	Average	5.50	Overwhelming	Balanced	Heavy	0.96
234	Low	Surplus	Loud	Engaged	Spotted	Expert	6.10	Overwhelming	Balanced	Heavy	1.00
235	Low	Surplus	Loud	Swarmed	Hidden	Novice	0.90	Passive	Generous	Mixed	1.00
236	Low	Surplus	Loud	Swarmed	Hidden	Average	1.50	Passive	Generous	Mixed	0.92
237	Low	Surplus	Loud	Swarmed	Hidden	Expert	2.10	Passive	Generous	Mixed	0.68
238	Low	Surplus	Loud	Swarmed	Suspicious	Novice	2.60	Tactical	Generous	Mixed	0.72
239	Low	Surplus	Loud	Swarmed	Suspicious	Average	3.20	Tactical	Generous	Mixed	0.96
240	Low	Surplus	Loud	Swarmed	Suspicious	Expert	3.80	Tactical	Generous	Heavy	0.92
241	Low	Surplus	Loud	Swarmed	Spotted	Novice	4.30	Tactical	Generous	Heavy	0.72
242	Low	Surplus	Loud	Swarmed	Spotted	Average	4.90	Overwhelming	Generous	Heavy	0.72
243	Low	Surplus	Loud	Swarmed	Spotted	Expert	5.50	Overwhelming	Generous	Heavy	0.96
244	Medium	Depleted	Quiet	Safe	Hidden	Novice	-0.05	Passive	Balanced	Swarm	1.00
245	Medium	Depleted	Quiet	Safe	Hidden	Average	0.55	Passive	Balanced	Swarm	1.00
246	Medium	Depleted	Quiet	Safe	Hidden	Expert	1.15	Passive	Balanced	Swarm	1.00
247	Medium	Depleted	Quiet	Safe	Suspicious	Novice	1.65	Passive	Balanced	Swarm	0.86
248	Medium	Depleted	Quiet	Safe	Suspicious	Average	2.25	Passive	Balanced	Swarm	0.62
249	Medium	Depleted	Quiet	Safe	Suspicious	Expert	2.85	Tactical	Balanced	Mixed	0.82
250	Medium	Depleted	Quiet	Safe	Spotted	Novice	3.35	Tactical	Balanced	Mixed	1.00
251	Medium	Depleted	Quiet	Safe	Spotted	Average	3.95	Tactical	Balanced	Mixed	0.86
252	Medium	Depleted	Quiet	Safe	Spotted	Expert	4.55	Tactical	Balanced	Mixed	0.62
253	Medium	Depleted	Quiet	Engaged	Hidden	Novice	-0.65	Passive	Generous	Swarm	1.00
254	Medium	Depleted	Quiet	Engaged	Hidden	Average	-0.05	Passive	Generous	Swarm	1.00
255	Medium	Depleted	Quiet	Engaged	Hidden	Expert	0.55	Passive	Generous	Swarm	1.00
256	Medium	Depleted	Quiet	Engaged	Suspicious	Novice	1.05	Passive	Generous	Swarm	1.00
257	Medium	Depleted	Quiet	Engaged	Suspicious	Average	1.65	Passive	Generous	Swarm	0.86
258	Medium	Depleted	Quiet	Engaged	Suspicious	Expert	2.25	Passive	Generous	Mixed	0.62
259	Medium	Depleted	Quiet	Engaged	Spotted	Novice	2.75	Tactical	Generous	Mixed	0.78
260	Medium	Depleted	Quiet	Engaged	Spotted	Average	3.35	Tactical	Generous	Mixed	1.00
261	Medium	Depleted	Quiet	Engaged	Spotted	Expert	3.95	Tactical	Generous	Mixed	0.86
262	Medium	Depleted	Quiet	Swarmed	Hidden	Novice	-1.25	Passive	Generous	Swarm	1.00
263	Medium	Depleted	Quiet	Swarmed	Hidden	Average	-0.65	Passive	Generous	Swarm	1.00
264	Medium	Depleted	Quiet	Swarmed	Hidden	Expert	-0.05	Passive	Generous	Swarm	1.00
265	Medium	Depleted	Quiet	Swarmed	Suspicious	Novice	0.45	Passive	Generous	Swarm	1.00
266	Medium	Depleted	Quiet	Swarmed	Suspicious	Average	1.05	Passive	Generous	Swarm	1.00
267	Medium	Depleted	Quiet	Swarmed	Suspicious	Expert	1.65	Passive	Generous	Mixed	0.86
268	Medium	Depleted	Quiet	Swarmed	Spotted	Novice	2.15	Passive	Generous	Mixed	0.66
269	Medium	Depleted	Quiet	Swarmed	Spotted	Average	2.75	Tactical	Generous	Mixed	0.78
270	Medium	Depleted	Quiet	Swarmed	Spotted	Expert	3.35	Tactical	Generous	Mixed	1.00
271	Medium	Depleted	Moderate	Safe	Hidden	Novice	0.75	Passive	Balanced	Swarm	1.00
272	Medium	Depleted	Moderate	Safe	Hidden	Average	1.35	Passive	Balanced	Swarm	0.98

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
273	Medium	Depleted	Moderate	Safe	Hidden	Expert	1.95	Passive	Balanced	Mixed	0.74
274	Medium	Depleted	Moderate	Safe	Suspicious	Novice	2.45	Tactical	Balanced	Swarm	0.66
275	Medium	Depleted	Moderate	Safe	Suspicious	Average	3.05	Tactical	Balanced	Mixed	0.90
276	Medium	Depleted	Moderate	Safe	Suspicious	Expert	3.65	Tactical	Balanced	Mixed	0.98
277	Medium	Depleted	Moderate	Safe	Spotted	Novice	4.15	Tactical	Balanced	Mixed	0.78
278	Medium	Depleted	Moderate	Safe	Spotted	Average	4.75	Overwhelming	Balanced	Mixed	0.66
279	Medium	Depleted	Moderate	Safe	Spotted	Expert	5.35	Overwhelming	Balanced	Mixed	0.90
280	Medium	Depleted	Moderate	Engaged	Hidden	Novice	0.15	Passive	Generous	Swarm	1.00
281	Medium	Depleted	Moderate	Engaged	Hidden	Average	0.75	Passive	Generous	Swarm	1.00
282	Medium	Depleted	Moderate	Engaged	Hidden	Expert	1.35	Passive	Generous	Mixed	0.98
283	Medium	Depleted	Moderate	Engaged	Suspicious	Novice	1.85	Passive	Generous	Swarm	0.78
284	Medium	Depleted	Moderate	Engaged	Suspicious	Average	2.45	Tactical	Generous	Mixed	0.66
285	Medium	Depleted	Moderate	Engaged	Suspicious	Expert	3.05	Tactical	Generous	Mixed	0.90
286	Medium	Depleted	Moderate	Engaged	Spotted	Novice	3.55	Tactical	Generous	Mixed	1.00
287	Medium	Depleted	Moderate	Engaged	Spotted	Average	4.15	Tactical	Generous	Mixed	0.78
288	Medium	Depleted	Moderate	Engaged	Spotted	Expert	4.75	Overwhelming	Generous	Mixed	0.66
289	Medium	Depleted	Moderate	Swarmed	Hidden	Novice	-0.45	Passive	Generous	Swarm	1.00
290	Medium	Depleted	Moderate	Swarmed	Hidden	Average	0.15	Passive	Generous	Swarm	1.00
291	Medium	Depleted	Moderate	Swarmed	Hidden	Expert	0.75	Passive	Generous	Mixed	1.00
292	Medium	Depleted	Moderate	Swarmed	Suspicious	Novice	1.25	Passive	Generous	Swarm	1.00
293	Medium	Depleted	Moderate	Swarmed	Suspicious	Average	1.85	Passive	Generous	Mixed	0.78
294	Medium	Depleted	Moderate	Swarmed	Suspicious	Expert	2.45	Tactical	Generous	Mixed	0.66
295	Medium	Depleted	Moderate	Swarmed	Spotted	Novice	2.95	Tactical	Generous	Mixed	0.86
296	Medium	Depleted	Moderate	Swarmed	Spotted	Average	3.55	Tactical	Generous	Mixed	1.00
297	Medium	Depleted	Moderate	Swarmed	Spotted	Expert	4.15	Tactical	Generous	Mixed	0.78
298	Medium	Depleted	Loud	Safe	Hidden	Novice	1.55	Passive	Balanced	Swarm	0.90
299	Medium	Depleted	Loud	Safe	Hidden	Average	2.15	Passive	Balanced	Mixed	0.66
300	Medium	Depleted	Loud	Safe	Hidden	Expert	2.75	Tactical	Balanced	Mixed	0.78
301	Medium	Depleted	Loud	Safe	Suspicious	Novice	3.25	Tactical	Balanced	Mixed	0.98
302	Medium	Depleted	Loud	Safe	Suspicious	Average	3.85	Tactical	Balanced	Mixed	0.90
303	Medium	Depleted	Loud	Safe	Suspicious	Expert	4.45	Tactical	Balanced	Mixed	0.66
304	Medium	Depleted	Loud	Safe	Spotted	Novice	4.95	Overwhelming	Balanced	Mixed	0.74
305	Medium	Depleted	Loud	Safe	Spotted	Average	5.55	Overwhelming	Balanced	Heavy	0.98
306	Medium	Depleted	Loud	Safe	Spotted	Expert	6.15	Overwhelming	Balanced	Heavy	1.00
307	Medium	Depleted	Loud	Engaged	Hidden	Novice	0.95	Passive	Generous	Swarm	1.00
308	Medium	Depleted	Loud	Engaged	Hidden	Average	1.55	Passive	Generous	Mixed	0.90
309	Medium	Depleted	Loud	Engaged	Hidden	Expert	2.15	Passive	Generous	Mixed	0.66
310	Medium	Depleted	Loud	Engaged	Suspicious	Novice	2.65	Tactical	Generous	Mixed	0.74
311	Medium	Depleted	Loud	Engaged	Suspicious	Average	3.25	Tactical	Generous	Mixed	0.98
312	Medium	Depleted	Loud	Engaged	Suspicious	Expert	3.85	Tactical	Generous	Mixed	0.90
313	Medium	Depleted	Loud	Engaged	Spotted	Novice	4.35	Tactical	Generous	Mixed	0.70
314	Medium	Depleted	Loud	Engaged	Spotted	Average	4.95	Overwhelming	Generous	Heavy	0.74
315	Medium	Depleted	Loud	Engaged	Spotted	Expert	5.55	Overwhelming	Generous	Heavy	0.98
316	Medium	Depleted	Loud	Swarmed	Hidden	Novice	0.35	Passive	Generous	Swarm	1.00
317	Medium	Depleted	Loud	Swarmed	Hidden	Average	0.95	Passive	Generous	Mixed	1.00
318	Medium	Depleted	Loud	Swarmed	Hidden	Expert	1.55	Passive	Generous	Mixed	0.90
319	Medium	Depleted	Loud	Swarmed	Suspicious	Novice	2.05	Passive	Generous	Mixed	0.70

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
320	Medium	Depleted	Loud	Swarmed	Suspicious	Average	2.65	Tactical	Generous	Mixed	0.74
321	Medium	Depleted	Loud	Swarmed	Suspicious	Expert	3.25	Tactical	Generous	Mixed	0.98
322	Medium	Depleted	Loud	Swarmed	Spotted	Novice	3.75	Tactical	Generous	Mixed	0.94
323	Medium	Depleted	Loud	Swarmed	Spotted	Average	4.35	Tactical	Generous	Heavy	0.70
324	Medium	Depleted	Loud	Swarmed	Spotted	Expert	4.95	Overwhelming	Generous	Heavy	0.74
325	Medium	Adequate	Quiet	Safe	Hidden	Novice	0.50	Passive	Balanced	Swarm	1.00
326	Medium	Adequate	Quiet	Safe	Hidden	Average	1.10	Passive	Balanced	Swarm	1.00
327	Medium	Adequate	Quiet	Safe	Hidden	Expert	1.70	Passive	Balanced	Swarm	0.84
328	Medium	Adequate	Quiet	Safe	Suspicious	Novice	2.20	Passive	Balanced	Swarm	0.64
329	Medium	Adequate	Quiet	Safe	Suspicious	Average	2.80	Tactical	Balanced	Mixed	0.80
330	Medium	Adequate	Quiet	Safe	Suspicious	Expert	3.40	Tactical	Balanced	Mixed	1.00
331	Medium	Adequate	Quiet	Safe	Spotted	Novice	3.90	Tactical	Balanced	Mixed	0.88
332	Medium	Adequate	Quiet	Safe	Spotted	Average	4.50	Tactical	Balanced	Mixed	0.64
333	Medium	Adequate	Quiet	Safe	Spotted	Expert	5.10	Overwhelming	Balanced	Mixed	0.80
334	Medium	Adequate	Quiet	Engaged	Hidden	Novice	-0.10	Passive	Balanced	Swarm	1.00
335	Medium	Adequate	Quiet	Engaged	Hidden	Average	0.50	Passive	Balanced	Swarm	1.00
336	Medium	Adequate	Quiet	Engaged	Hidden	Expert	1.10	Passive	Balanced	Swarm	1.00
337	Medium	Adequate	Quiet	Engaged	Suspicious	Novice	1.60	Passive	Balanced	Swarm	0.88
338	Medium	Adequate	Quiet	Engaged	Suspicious	Average	2.20	Passive	Balanced	Mixed	0.64
339	Medium	Adequate	Quiet	Engaged	Suspicious	Expert	2.80	Tactical	Balanced	Mixed	0.80
340	Medium	Adequate	Quiet	Engaged	Spotted	Novice	3.30	Tactical	Balanced	Mixed	1.00
341	Medium	Adequate	Quiet	Engaged	Spotted	Average	3.90	Tactical	Balanced	Mixed	0.88
342	Medium	Adequate	Quiet	Engaged	Spotted	Expert	4.50	Tactical	Balanced	Mixed	0.64
343	Medium	Adequate	Quiet	Swarmed	Hidden	Novice	-0.70	Passive	Generous	Swarm	1.00
344	Medium	Adequate	Quiet	Swarmed	Hidden	Average	-0.10	Passive	Generous	Swarm	1.00
345	Medium	Adequate	Quiet	Swarmed	Hidden	Expert	0.50	Passive	Generous	Swarm	1.00
346	Medium	Adequate	Quiet	Swarmed	Suspicious	Novice	1.00	Passive	Generous	Swarm	1.00
347	Medium	Adequate	Quiet	Swarmed	Suspicious	Average	1.60	Passive	Generous	Mixed	0.88
348	Medium	Adequate	Quiet	Swarmed	Suspicious	Expert	2.20	Passive	Generous	Mixed	0.64
349	Medium	Adequate	Quiet	Swarmed	Spotted	Novice	2.70	Tactical	Generous	Mixed	0.76
350	Medium	Adequate	Quiet	Swarmed	Spotted	Average	3.30	Tactical	Generous	Mixed	1.00
351	Medium	Adequate	Quiet	Swarmed	Spotted	Expert	3.90	Tactical	Generous	Mixed	0.88
352	Medium	Adequate	Moderate	Safe	Hidden	Novice	1.30	Passive	Balanced	Swarm	1.00
353	Medium	Adequate	Moderate	Safe	Hidden	Average	1.90	Passive	Balanced	Mixed	0.76
354	Medium	Adequate	Moderate	Safe	Hidden	Expert	2.50	Tactical	Balanced	Mixed	0.68
355	Medium	Adequate	Moderate	Safe	Suspicious	Novice	3.00	Tactical	Balanced	Mixed	0.88
356	Medium	Adequate	Moderate	Safe	Suspicious	Average	3.60	Tactical	Balanced	Mixed	1.00
357	Medium	Adequate	Moderate	Safe	Suspicious	Expert	4.20	Tactical	Balanced	Mixed	0.76
358	Medium	Adequate	Moderate	Safe	Spotted	Novice	4.70	Overwhelming	Balanced	Mixed	0.64
359	Medium	Adequate	Moderate	Safe	Spotted	Average	5.30	Overwhelming	Balanced	Mixed	0.88
360	Medium	Adequate	Moderate	Safe	Spotted	Expert	5.90	Overwhelming	Balanced	Heavy	1.00
361	Medium	Adequate	Moderate	Engaged	Hidden	Novice	0.70	Passive	Balanced	Swarm	1.00
362	Medium	Adequate	Moderate	Engaged	Hidden	Average	1.30	Passive	Balanced	Mixed	1.00
363	Medium	Adequate	Moderate	Engaged	Hidden	Expert	1.90	Passive	Balanced	Mixed	0.76
364	Medium	Adequate	Moderate	Engaged	Suspicious	Novice	2.40	Tactical	Balanced	Mixed	0.64
365	Medium	Adequate	Moderate	Engaged	Suspicious	Average	3.00	Tactical	Balanced	Mixed	0.88
366	Medium	Adequate	Moderate	Engaged	Suspicious	Expert	3.60	Tactical	Balanced	Mixed	1.00

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
367	Medium	Adequate	Moderate	Engaged	Spotted	Novice	4.10	Tactical	Balanced	Mixed	0.80
368	Medium	Adequate	Moderate	Engaged	Spotted	Average	4.70	Overwhelming	Balanced	Mixed	0.64
369	Medium	Adequate	Moderate	Engaged	Spotted	Expert	5.30	Overwhelming	Balanced	Heavy	0.88
370	Medium	Adequate	Moderate	Swarmed	Hidden	Novice	0.10	Passive	Generous	Swarm	1.00
371	Medium	Adequate	Moderate	Swarmed	Hidden	Average	0.70	Passive	Generous	Mixed	1.00
372	Medium	Adequate	Moderate	Swarmed	Hidden	Expert	1.30	Passive	Generous	Mixed	1.00
373	Medium	Adequate	Moderate	Swarmed	Suspicious	Novice	1.80	Passive	Generous	Mixed	0.80
374	Medium	Adequate	Moderate	Swarmed	Suspicious	Average	2.40	Tactical	Generous	Mixed	0.64
375	Medium	Adequate	Moderate	Swarmed	Suspicious	Expert	3.00	Tactical	Generous	Mixed	0.88
376	Medium	Adequate	Moderate	Swarmed	Spotted	Novice	3.50	Tactical	Generous	Mixed	1.00
377	Medium	Adequate	Moderate	Swarmed	Spotted	Average	4.10	Tactical	Generous	Mixed	0.80
378	Medium	Adequate	Moderate	Swarmed	Spotted	Expert	4.70	Overwhelming	Generous	Heavy	0.64
379	Medium	Adequate	Loud	Safe	Hidden	Novice	2.10	Passive	Balanced	Mixed	0.68
380	Medium	Adequate	Loud	Safe	Hidden	Average	2.70	Tactical	Balanced	Mixed	0.76
381	Medium	Adequate	Loud	Safe	Hidden	Expert	3.30	Tactical	Balanced	Mixed	1.00
382	Medium	Adequate	Loud	Safe	Suspicious	Novice	3.80	Tactical	Balanced	Mixed	0.92
383	Medium	Adequate	Loud	Safe	Suspicious	Average	4.40	Tactical	Balanced	Mixed	0.68
384	Medium	Adequate	Loud	Safe	Suspicious	Expert	5.00	Overwhelming	Balanced	Heavy	0.76
385	Medium	Adequate	Loud	Safe	Spotted	Novice	5.50	Overwhelming	Balanced	Heavy	0.96
386	Medium	Adequate	Loud	Safe	Spotted	Average	6.10	Overwhelming	Balanced	Heavy	1.00
387	Medium	Adequate	Loud	Safe	Spotted	Expert	6.70	Overwhelming	Balanced	Heavy	1.00
388	Medium	Adequate	Loud	Engaged	Hidden	Novice	1.50	Passive	Balanced	Mixed	0.92
389	Medium	Adequate	Loud	Engaged	Hidden	Average	2.10	Passive	Balanced	Mixed	0.68
390	Medium	Adequate	Loud	Engaged	Hidden	Expert	2.70	Tactical	Balanced	Mixed	0.76
391	Medium	Adequate	Loud	Engaged	Suspicious	Novice	3.20	Tactical	Balanced	Mixed	0.96
392	Medium	Adequate	Loud	Engaged	Suspicious	Average	3.80	Tactical	Balanced	Mixed	0.92
393	Medium	Adequate	Loud	Engaged	Suspicious	Expert	4.40	Tactical	Balanced	Heavy	0.68
394	Medium	Adequate	Loud	Engaged	Spotted	Novice	4.90	Overwhelming	Balanced	Heavy	0.72
395	Medium	Adequate	Loud	Engaged	Spotted	Average	5.50	Overwhelming	Balanced	Heavy	0.96
396	Medium	Adequate	Loud	Engaged	Spotted	Expert	6.10	Overwhelming	Balanced	Heavy	1.00
397	Medium	Adequate	Loud	Swarmed	Hidden	Novice	0.90	Passive	Generous	Mixed	1.00
398	Medium	Adequate	Loud	Swarmed	Hidden	Average	1.50	Passive	Generous	Mixed	0.92
399	Medium	Adequate	Loud	Swarmed	Hidden	Expert	2.10	Passive	Generous	Mixed	0.68
400	Medium	Adequate	Loud	Swarmed	Suspicious	Novice	2.60	Tactical	Generous	Mixed	0.72
401	Medium	Adequate	Loud	Swarmed	Suspicious	Average	3.20	Tactical	Generous	Mixed	0.96
402	Medium	Adequate	Loud	Swarmed	Suspicious	Expert	3.80	Tactical	Generous	Heavy	0.92
403	Medium	Adequate	Loud	Swarmed	Spotted	Novice	4.30	Tactical	Generous	Heavy	0.72
404	Medium	Adequate	Loud	Swarmed	Spotted	Average	4.90	Overwhelming	Generous	Heavy	0.72
405	Medium	Adequate	Loud	Swarmed	Spotted	Expert	5.50	Overwhelming	Generous	Heavy	0.96
406	Medium	Surplus	Quiet	Safe	Hidden	Novice	1.05	Passive	Scarce	Swarm	1.00
407	Medium	Surplus	Quiet	Safe	Hidden	Average	1.65	Passive	Scarce	Swarm	0.86
408	Medium	Surplus	Quiet	Safe	Hidden	Expert	2.25	Passive	Scarce	Mixed	0.62
409	Medium	Surplus	Quiet	Safe	Suspicious	Novice	2.75	Tactical	Scarce	Mixed	0.78
410	Medium	Surplus	Quiet	Safe	Suspicious	Average	3.35	Tactical	Scarce	Mixed	1.00
411	Medium	Surplus	Quiet	Safe	Suspicious	Expert	3.95	Tactical	Scarce	Mixed	0.86
412	Medium	Surplus	Quiet	Safe	Spotted	Novice	4.45	Tactical	Scarce	Mixed	0.66
413	Medium	Surplus	Quiet	Safe	Spotted	Average	5.05	Overwhelming	Scarce	Mixed	0.78

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
414	Medium	Surplus	Quiet	Safe	Spotted	Expert	5.65	Overwhelming	Scarce	Heavy	1.00
415	Medium	Surplus	Quiet	Engaged	Hidden	Novice	0.45	Passive	Scarce	Swarm	1.00
416	Medium	Surplus	Quiet	Engaged	Hidden	Average	1.05	Passive	Scarce	Swarm	1.00
417	Medium	Surplus	Quiet	Engaged	Hidden	Expert	1.65	Passive	Scarce	Mixed	0.86
418	Medium	Surplus	Quiet	Engaged	Suspicious	Novice	2.15	Passive	Scarce	Mixed	0.66
419	Medium	Surplus	Quiet	Engaged	Suspicious	Average	2.75	Tactical	Scarce	Mixed	0.78
420	Medium	Surplus	Quiet	Engaged	Suspicious	Expert	3.35	Tactical	Scarce	Mixed	1.00
421	Medium	Surplus	Quiet	Engaged	Spotted	Novice	3.85	Tactical	Scarce	Mixed	0.90
422	Medium	Surplus	Quiet	Engaged	Spotted	Average	4.45	Tactical	Scarce	Mixed	0.66
423	Medium	Surplus	Quiet	Engaged	Spotted	Expert	5.05	Overwhelming	Scarce	Heavy	0.78
424	Medium	Surplus	Quiet	Swarmed	Hidden	Novice	-0.15	Passive	Balanced	Swarm	1.00
425	Medium	Surplus	Quiet	Swarmed	Hidden	Average	0.45	Passive	Balanced	Swarm	1.00
426	Medium	Surplus	Quiet	Swarmed	Hidden	Expert	1.05	Passive	Balanced	Mixed	1.00
427	Medium	Surplus	Quiet	Swarmed	Suspicious	Novice	1.55	Passive	Balanced	Mixed	0.90
428	Medium	Surplus	Quiet	Swarmed	Suspicious	Average	2.15	Passive	Balanced	Mixed	0.66
429	Medium	Surplus	Quiet	Swarmed	Suspicious	Expert	2.75	Tactical	Balanced	Mixed	0.78
430	Medium	Surplus	Quiet	Swarmed	Spotted	Novice	3.25	Tactical	Balanced	Mixed	0.98
431	Medium	Surplus	Quiet	Swarmed	Spotted	Average	3.85	Tactical	Balanced	Mixed	0.90
432	Medium	Surplus	Quiet	Swarmed	Spotted	Expert	4.45	Tactical	Balanced	Heavy	0.66
433	Medium	Surplus	Moderate	Safe	Hidden	Novice	1.85	Passive	Scarce	Mixed	0.78
434	Medium	Surplus	Moderate	Safe	Hidden	Average	2.45	Tactical	Scarce	Mixed	0.66
435	Medium	Surplus	Moderate	Safe	Hidden	Expert	3.05	Tactical	Scarce	Mixed	0.90
436	Medium	Surplus	Moderate	Safe	Suspicious	Novice	3.55	Tactical	Scarce	Mixed	1.00
437	Medium	Surplus	Moderate	Safe	Suspicious	Average	4.15	Tactical	Scarce	Mixed	0.78
438	Medium	Surplus	Moderate	Safe	Suspicious	Expert	4.75	Overwhelming	Scarce	Heavy	0.66
439	Medium	Surplus	Moderate	Safe	Spotted	Novice	5.25	Overwhelming	Scarce	Mixed	0.86
440	Medium	Surplus	Moderate	Safe	Spotted	Average	5.85	Overwhelming	Scarce	Heavy	1.00
441	Medium	Surplus	Moderate	Safe	Spotted	Expert	6.45	Overwhelming	Scarce	Heavy	1.00
442	Medium	Surplus	Moderate	Engaged	Hidden	Novice	1.25	Passive	Scarce	Mixed	1.00
443	Medium	Surplus	Moderate	Engaged	Hidden	Average	1.85	Passive	Scarce	Mixed	0.78
444	Medium	Surplus	Moderate	Engaged	Hidden	Expert	2.45	Tactical	Scarce	Mixed	0.66
445	Medium	Surplus	Moderate	Engaged	Suspicious	Novice	2.95	Tactical	Scarce	Mixed	0.86
446	Medium	Surplus	Moderate	Engaged	Suspicious	Average	3.55	Tactical	Scarce	Mixed	1.00
447	Medium	Surplus	Moderate	Engaged	Suspicious	Expert	4.15	Tactical	Scarce	Heavy	0.78
448	Medium	Surplus	Moderate	Engaged	Spotted	Novice	4.65	Overwhelming	Scarce	Mixed	0.62
449	Medium	Surplus	Moderate	Engaged	Spotted	Average	5.25	Overwhelming	Scarce	Heavy	0.86
450	Medium	Surplus	Moderate	Engaged	Spotted	Expert	5.85	Overwhelming	Scarce	Heavy	1.00
451	Medium	Surplus	Moderate	Swarmed	Hidden	Novice	0.65	Passive	Balanced	Mixed	1.00
452	Medium	Surplus	Moderate	Swarmed	Hidden	Average	1.25	Passive	Balanced	Mixed	1.00
453	Medium	Surplus	Moderate	Swarmed	Hidden	Expert	1.85	Passive	Balanced	Mixed	0.78
454	Medium	Surplus	Moderate	Swarmed	Suspicious	Novice	2.35	Tactical	Balanced	Mixed	0.62
455	Medium	Surplus	Moderate	Swarmed	Suspicious	Average	2.95	Tactical	Balanced	Mixed	0.86
456	Medium	Surplus	Moderate	Swarmed	Suspicious	Expert	3.55	Tactical	Balanced	Heavy	1.00
457	Medium	Surplus	Moderate	Swarmed	Spotted	Novice	4.05	Tactical	Balanced	Mixed	0.82
458	Medium	Surplus	Moderate	Swarmed	Spotted	Average	4.65	Overwhelming	Balanced	Heavy	0.62
459	Medium	Surplus	Moderate	Swarmed	Spotted	Expert	5.25	Overwhelming	Balanced	Heavy	0.86
460	Medium	Surplus	Loud	Safe	Hidden	Novice	2.65	Tactical	Scarce	Mixed	0.74

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
461	Medium	Surplus	Loud	Safe	Hidden	Average	3.25	Tactical	Scarce	Mixed	0.98
462	Medium	Surplus	Loud	Safe	Hidden	Expert	3.85	Tactical	Scarce	Heavy	0.90
463	Medium	Surplus	Loud	Safe	Suspicious	Novice	4.35	Tactical	Scarce	Mixed	0.70
464	Medium	Surplus	Loud	Safe	Suspicious	Average	4.95	Overwhelming	Scarce	Heavy	0.74
465	Medium	Surplus	Loud	Safe	Suspicious	Expert	5.55	Overwhelming	Scarce	Heavy	0.98
466	Medium	Surplus	Loud	Safe	Spotted	Novice	6.05	Overwhelming	Scarce	Heavy	1.00
467	Medium	Surplus	Loud	Safe	Spotted	Average	6.65	Overwhelming	Scarce	Heavy	1.00
468	Medium	Surplus	Loud	Safe	Spotted	Expert	7.25	Overwhelming	Scarce	Heavy	1.00
469	Medium	Surplus	Loud	Engaged	Hidden	Novice	2.05	Passive	Scarce	Mixed	0.70
470	Medium	Surplus	Loud	Engaged	Hidden	Average	2.65	Tactical	Scarce	Mixed	0.74
471	Medium	Surplus	Loud	Engaged	Hidden	Expert	3.25	Tactical	Scarce	Heavy	0.98
472	Medium	Surplus	Loud	Engaged	Suspicious	Novice	3.75	Tactical	Scarce	Mixed	0.94
473	Medium	Surplus	Loud	Engaged	Suspicious	Average	4.35	Tactical	Scarce	Heavy	0.70
474	Medium	Surplus	Loud	Engaged	Suspicious	Expert	4.95	Overwhelming	Scarce	Heavy	0.74
475	Medium	Surplus	Loud	Engaged	Spotted	Novice	5.45	Overwhelming	Scarce	Heavy	0.94
476	Medium	Surplus	Loud	Engaged	Spotted	Average	6.05	Overwhelming	Scarce	Heavy	1.00
477	Medium	Surplus	Loud	Engaged	Spotted	Expert	6.65	Overwhelming	Scarce	Heavy	1.00
478	Medium	Surplus	Loud	Swarmed	Hidden	Novice	1.45	Passive	Balanced	Mixed	0.94
479	Medium	Surplus	Loud	Swarmed	Hidden	Average	2.05	Passive	Balanced	Mixed	0.70
480	Medium	Surplus	Loud	Swarmed	Hidden	Expert	2.65	Tactical	Balanced	Heavy	0.74
481	Medium	Surplus	Loud	Swarmed	Suspicious	Novice	3.15	Tactical	Balanced	Mixed	0.94
482	Medium	Surplus	Loud	Swarmed	Suspicious	Average	3.75	Tactical	Balanced	Heavy	0.94
483	Medium	Surplus	Loud	Swarmed	Suspicious	Expert	4.35	Tactical	Balanced	Heavy	0.70
484	Medium	Surplus	Loud	Swarmed	Spotted	Novice	4.85	Overwhelming	Balanced	Heavy	0.70
485	Medium	Surplus	Loud	Swarmed	Spotted	Average	5.45	Overwhelming	Balanced	Heavy	0.94
486	Medium	Surplus	Loud	Swarmed	Spotted	Expert	6.05	Overwhelming	Balanced	Heavy	1.00
487	High	Depleted	Quiet	Safe	Hidden	Novice	0.50	Passive	Balanced	Swarm	1.00
488	High	Depleted	Quiet	Safe	Hidden	Average	1.10	Passive	Balanced	Swarm	1.00
489	High	Depleted	Quiet	Safe	Hidden	Expert	1.70	Passive	Balanced	Swarm	0.84
490	High	Depleted	Quiet	Safe	Suspicious	Novice	2.20	Passive	Balanced	Swarm	0.64
491	High	Depleted	Quiet	Safe	Suspicious	Average	2.80	Tactical	Balanced	Mixed	0.80
492	High	Depleted	Quiet	Safe	Suspicious	Expert	3.40	Tactical	Balanced	Mixed	1.00
493	High	Depleted	Quiet	Safe	Spotted	Novice	3.90	Tactical	Balanced	Mixed	0.88
494	High	Depleted	Quiet	Safe	Spotted	Average	4.50	Tactical	Balanced	Mixed	0.64
495	High	Depleted	Quiet	Safe	Spotted	Expert	5.10	Overwhelming	Balanced	Mixed	0.80
496	High	Depleted	Quiet	Engaged	Hidden	Novice	-0.10	Passive	Balanced	Swarm	1.00
497	High	Depleted	Quiet	Engaged	Hidden	Average	0.50	Passive	Balanced	Swarm	1.00
498	High	Depleted	Quiet	Engaged	Hidden	Expert	1.10	Passive	Balanced	Swarm	1.00
499	High	Depleted	Quiet	Engaged	Suspicious	Novice	1.60	Passive	Balanced	Swarm	0.88
500	High	Depleted	Quiet	Engaged	Suspicious	Average	2.20	Passive	Balanced	Mixed	0.64
501	High	Depleted	Quiet	Engaged	Suspicious	Expert	2.80	Tactical	Balanced	Mixed	0.80
502	High	Depleted	Quiet	Engaged	Spotted	Novice	3.30	Tactical	Balanced	Mixed	1.00
503	High	Depleted	Quiet	Engaged	Spotted	Average	3.90	Tactical	Balanced	Mixed	0.88
504	High	Depleted	Quiet	Engaged	Spotted	Expert	4.50	Tactical	Balanced	Mixed	0.64
505	High	Depleted	Quiet	Swarmed	Hidden	Novice	-0.70	Passive	Generous	Swarm	1.00
506	High	Depleted	Quiet	Swarmed	Hidden	Average	-0.10	Passive	Generous	Swarm	1.00
507	High	Depleted	Quiet	Swarmed	Hidden	Expert	0.50	Passive	Generous	Swarm	1.00

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
508	High	Depleted	Quiet	Swarmed	Suspicious	Novice	1.00	Passive	Generous	Swarm	1.00
509	High	Depleted	Quiet	Swarmed	Suspicious	Average	1.60	Passive	Generous	Mixed	0.88
510	High	Depleted	Quiet	Swarmed	Suspicious	Expert	2.20	Passive	Generous	Mixed	0.64
511	High	Depleted	Quiet	Swarmed	Spotted	Novice	2.70	Tactical	Generous	Mixed	0.76
512	High	Depleted	Quiet	Swarmed	Spotted	Average	3.30	Tactical	Generous	Mixed	1.00
513	High	Depleted	Quiet	Swarmed	Spotted	Expert	3.90	Tactical	Generous	Mixed	0.88
514	High	Depleted	Moderate	Safe	Hidden	Novice	1.30	Passive	Balanced	Swarm	1.00
515	High	Depleted	Moderate	Safe	Hidden	Average	1.90	Passive	Balanced	Mixed	0.76
516	High	Depleted	Moderate	Safe	Hidden	Expert	2.50	Tactical	Balanced	Mixed	0.68
517	High	Depleted	Moderate	Safe	Suspicious	Novice	3.00	Tactical	Balanced	Mixed	0.88
518	High	Depleted	Moderate	Safe	Suspicious	Average	3.60	Tactical	Balanced	Mixed	1.00
519	High	Depleted	Moderate	Safe	Suspicious	Expert	4.20	Tactical	Balanced	Mixed	0.76
520	High	Depleted	Moderate	Safe	Spotted	Novice	4.70	Overwhelming	Balanced	Mixed	0.64
521	High	Depleted	Moderate	Safe	Spotted	Average	5.30	Overwhelming	Balanced	Mixed	0.88
522	High	Depleted	Moderate	Safe	Spotted	Expert	5.90	Overwhelming	Balanced	Heavy	1.00
523	High	Depleted	Moderate	Engaged	Hidden	Novice	0.70	Passive	Balanced	Swarm	1.00
524	High	Depleted	Moderate	Engaged	Hidden	Average	1.30	Passive	Balanced	Mixed	1.00
525	High	Depleted	Moderate	Engaged	Hidden	Expert	1.90	Passive	Balanced	Mixed	0.76
526	High	Depleted	Moderate	Engaged	Suspicious	Novice	2.40	Tactical	Balanced	Mixed	0.64
527	High	Depleted	Moderate	Engaged	Suspicious	Average	3.00	Tactical	Balanced	Mixed	0.88
528	High	Depleted	Moderate	Engaged	Suspicious	Expert	3.60	Tactical	Balanced	Mixed	1.00
529	High	Depleted	Moderate	Engaged	Spotted	Novice	4.10	Tactical	Balanced	Mixed	0.80
530	High	Depleted	Moderate	Engaged	Spotted	Average	4.70	Overwhelming	Balanced	Mixed	0.64
531	High	Depleted	Moderate	Engaged	Spotted	Expert	5.30	Overwhelming	Balanced	Heavy	0.88
532	High	Depleted	Moderate	Swarmed	Hidden	Novice	0.10	Passive	Generous	Swarm	1.00
533	High	Depleted	Moderate	Swarmed	Hidden	Average	0.70	Passive	Generous	Mixed	1.00
534	High	Depleted	Moderate	Swarmed	Hidden	Expert	1.30	Passive	Generous	Mixed	1.00
535	High	Depleted	Moderate	Swarmed	Suspicious	Novice	1.80	Passive	Generous	Mixed	0.80
536	High	Depleted	Moderate	Swarmed	Suspicious	Average	2.40	Tactical	Generous	Mixed	0.64
537	High	Depleted	Moderate	Swarmed	Suspicious	Expert	3.00	Tactical	Generous	Mixed	0.88
538	High	Depleted	Moderate	Swarmed	Spotted	Novice	3.50	Tactical	Generous	Mixed	1.00
539	High	Depleted	Moderate	Swarmed	Spotted	Average	4.10	Tactical	Generous	Mixed	0.80
540	High	Depleted	Moderate	Swarmed	Spotted	Expert	4.70	Overwhelming	Generous	Heavy	0.64
541	High	Depleted	Loud	Safe	Hidden	Novice	2.10	Passive	Balanced	Mixed	0.68
542	High	Depleted	Loud	Safe	Hidden	Average	2.70	Tactical	Balanced	Mixed	0.76
543	High	Depleted	Loud	Safe	Hidden	Expert	3.30	Tactical	Balanced	Mixed	1.00
544	High	Depleted	Loud	Safe	Suspicious	Novice	3.80	Tactical	Balanced	Mixed	0.92
545	High	Depleted	Loud	Safe	Suspicious	Average	4.40	Tactical	Balanced	Mixed	0.68
546	High	Depleted	Loud	Safe	Suspicious	Expert	5.00	Overwhelming	Balanced	Heavy	0.76
547	High	Depleted	Loud	Safe	Spotted	Novice	5.50	Overwhelming	Balanced	Heavy	0.96
548	High	Depleted	Loud	Safe	Spotted	Average	6.10	Overwhelming	Balanced	Heavy	1.00
549	High	Depleted	Loud	Safe	Spotted	Expert	6.70	Overwhelming	Balanced	Heavy	1.00
550	High	Depleted	Loud	Engaged	Hidden	Novice	1.50	Passive	Balanced	Mixed	0.92
551	High	Depleted	Loud	Engaged	Hidden	Average	2.10	Passive	Balanced	Mixed	0.68
552	High	Depleted	Loud	Engaged	Hidden	Expert	2.70	Tactical	Balanced	Mixed	0.76
553	High	Depleted	Loud	Engaged	Suspicious	Novice	3.20	Tactical	Balanced	Mixed	0.96
554	High	Depleted	Loud	Engaged	Suspicious	Average	3.80	Tactical	Balanced	Mixed	0.92

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
555	High	Depleted	Loud	Engaged	Suspicious	Expert	4.40	Tactical	Balanced	Heavy	0.68
556	High	Depleted	Loud	Engaged	Spotted	Novice	4.90	Overwhelming	Balanced	Heavy	0.72
557	High	Depleted	Loud	Engaged	Spotted	Average	5.50	Overwhelming	Balanced	Heavy	0.96
558	High	Depleted	Loud	Engaged	Spotted	Expert	6.10	Overwhelming	Balanced	Heavy	1.00
559	High	Depleted	Loud	Swarmed	Hidden	Novice	0.90	Passive	Generous	Mixed	1.00
560	High	Depleted	Loud	Swarmed	Hidden	Average	1.50	Passive	Generous	Mixed	0.92
561	High	Depleted	Loud	Swarmed	Hidden	Expert	2.10	Passive	Generous	Mixed	0.68
562	High	Depleted	Loud	Swarmed	Suspicious	Novice	2.60	Tactical	Generous	Mixed	0.72
563	High	Depleted	Loud	Swarmed	Suspicious	Average	3.20	Tactical	Generous	Mixed	0.96
564	High	Depleted	Loud	Swarmed	Suspicious	Expert	3.80	Tactical	Generous	Heavy	0.92
565	High	Depleted	Loud	Swarmed	Spotted	Novice	4.30	Tactical	Generous	Heavy	0.72
566	High	Depleted	Loud	Swarmed	Spotted	Average	4.90	Overwhelming	Generous	Heavy	0.72
567	High	Depleted	Loud	Swarmed	Spotted	Expert	5.50	Overwhelming	Generous	Heavy	0.96
568	High	Adequate	Quiet	Safe	Hidden	Novice	1.05	Passive	Scarce	Swarm	1.00
569	High	Adequate	Quiet	Safe	Hidden	Average	1.65	Passive	Scarce	Swarm	0.86
570	High	Adequate	Quiet	Safe	Hidden	Expert	2.25	Passive	Scarce	Mixed	0.62
571	High	Adequate	Quiet	Safe	Suspicious	Novice	2.75	Tactical	Scarce	Mixed	0.78
572	High	Adequate	Quiet	Safe	Suspicious	Average	3.35	Tactical	Scarce	Mixed	1.00
573	High	Adequate	Quiet	Safe	Suspicious	Expert	3.95	Tactical	Scarce	Mixed	0.86
574	High	Adequate	Quiet	Safe	Spotted	Novice	4.45	Tactical	Scarce	Mixed	0.66
575	High	Adequate	Quiet	Safe	Spotted	Average	5.05	Overwhelming	Scarce	Mixed	0.78
576	High	Adequate	Quiet	Safe	Spotted	Expert	5.65	Overwhelming	Scarce	Heavy	1.00
577	High	Adequate	Quiet	Engaged	Hidden	Novice	0.45	Passive	Scarce	Swarm	1.00
578	High	Adequate	Quiet	Engaged	Hidden	Average	1.05	Passive	Scarce	Swarm	1.00
579	High	Adequate	Quiet	Engaged	Hidden	Expert	1.65	Passive	Scarce	Mixed	0.86
580	High	Adequate	Quiet	Engaged	Suspicious	Novice	2.15	Passive	Scarce	Mixed	0.66
581	High	Adequate	Quiet	Engaged	Suspicious	Average	2.75	Tactical	Scarce	Mixed	0.78
582	High	Adequate	Quiet	Engaged	Suspicious	Expert	3.35	Tactical	Scarce	Mixed	1.00
583	High	Adequate	Quiet	Engaged	Spotted	Novice	3.85	Tactical	Scarce	Mixed	0.90
584	High	Adequate	Quiet	Engaged	Spotted	Average	4.45	Tactical	Scarce	Mixed	0.66
585	High	Adequate	Quiet	Engaged	Spotted	Expert	5.05	Overwhelming	Scarce	Heavy	0.78
586	High	Adequate	Quiet	Swarmed	Hidden	Novice	-0.15	Passive	Balanced	Swarm	1.00
587	High	Adequate	Quiet	Swarmed	Hidden	Average	0.45	Passive	Balanced	Swarm	1.00
588	High	Adequate	Quiet	Swarmed	Hidden	Expert	1.05	Passive	Balanced	Mixed	1.00
589	High	Adequate	Quiet	Swarmed	Suspicious	Novice	1.55	Passive	Balanced	Mixed	0.90
590	High	Adequate	Quiet	Swarmed	Suspicious	Average	2.15	Passive	Balanced	Mixed	0.66
591	High	Adequate	Quiet	Swarmed	Suspicious	Expert	2.75	Tactical	Balanced	Mixed	0.78
592	High	Adequate	Quiet	Swarmed	Spotted	Novice	3.25	Tactical	Balanced	Mixed	0.98
593	High	Adequate	Quiet	Swarmed	Spotted	Average	3.85	Tactical	Balanced	Mixed	0.90
594	High	Adequate	Quiet	Swarmed	Spotted	Expert	4.45	Tactical	Balanced	Heavy	0.66
595	High	Adequate	Moderate	Safe	Hidden	Novice	1.85	Passive	Scarce	Mixed	0.78
596	High	Adequate	Moderate	Safe	Hidden	Average	2.45	Tactical	Scarce	Mixed	0.66
597	High	Adequate	Moderate	Safe	Hidden	Expert	3.05	Tactical	Scarce	Mixed	0.90
598	High	Adequate	Moderate	Safe	Suspicious	Novice	3.55	Tactical	Scarce	Mixed	1.00
599	High	Adequate	Moderate	Safe	Suspicious	Average	4.15	Tactical	Scarce	Mixed	0.78
600	High	Adequate	Moderate	Safe	Suspicious	Expert	4.75	Overwhelming	Scarce	Heavy	0.66
601	High	Adequate	Moderate	Safe	Spotted	Novice	5.25	Overwhelming	Scarce	Mixed	0.86

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
602	High	Adequate	Moderate	Safe	Spotted	Average	5.85	Overwhelming	Scarce	Heavy	1.00
603	High	Adequate	Moderate	Safe	Spotted	Expert	6.45	Overwhelming	Scarce	Heavy	1.00
604	High	Adequate	Moderate	Engaged	Hidden	Novice	1.25	Passive	Scarce	Mixed	1.00
605	High	Adequate	Moderate	Engaged	Hidden	Average	1.85	Passive	Scarce	Mixed	0.78
606	High	Adequate	Moderate	Engaged	Hidden	Expert	2.45	Tactical	Scarce	Mixed	0.66
607	High	Adequate	Moderate	Engaged	Suspicious	Novice	2.95	Tactical	Scarce	Mixed	0.86
608	High	Adequate	Moderate	Engaged	Suspicious	Average	3.55	Tactical	Scarce	Mixed	1.00
609	High	Adequate	Moderate	Engaged	Suspicious	Expert	4.15	Tactical	Scarce	Heavy	0.78
610	High	Adequate	Moderate	Engaged	Spotted	Novice	4.65	Overwhelming	Scarce	Mixed	0.62
611	High	Adequate	Moderate	Engaged	Spotted	Average	5.25	Overwhelming	Scarce	Heavy	0.86
612	High	Adequate	Moderate	Engaged	Spotted	Expert	5.85	Overwhelming	Scarce	Heavy	1.00
613	High	Adequate	Moderate	Swarmed	Hidden	Novice	0.65	Passive	Balanced	Mixed	1.00
614	High	Adequate	Moderate	Swarmed	Hidden	Average	1.25	Passive	Balanced	Mixed	1.00
615	High	Adequate	Moderate	Swarmed	Hidden	Expert	1.85	Passive	Balanced	Mixed	0.78
616	High	Adequate	Moderate	Swarmed	Suspicious	Novice	2.35	Tactical	Balanced	Mixed	0.62
617	High	Adequate	Moderate	Swarmed	Suspicious	Average	2.95	Tactical	Balanced	Mixed	0.86
618	High	Adequate	Moderate	Swarmed	Suspicious	Expert	3.55	Tactical	Balanced	Heavy	1.00
619	High	Adequate	Moderate	Swarmed	Spotted	Novice	4.05	Tactical	Balanced	Mixed	0.82
620	High	Adequate	Moderate	Swarmed	Spotted	Average	4.65	Overwhelming	Balanced	Heavy	0.62
621	High	Adequate	Moderate	Swarmed	Spotted	Expert	5.25	Overwhelming	Balanced	Heavy	0.86
622	High	Adequate	Loud	Safe	Hidden	Novice	2.65	Tactical	Scarce	Mixed	0.74
623	High	Adequate	Loud	Safe	Hidden	Average	3.25	Tactical	Scarce	Mixed	0.98
624	High	Adequate	Loud	Safe	Hidden	Expert	3.85	Tactical	Scarce	Heavy	0.90
625	High	Adequate	Loud	Safe	Suspicious	Novice	4.35	Tactical	Scarce	Mixed	0.70
626	High	Adequate	Loud	Safe	Suspicious	Average	4.95	Overwhelming	Scarce	Heavy	0.74
627	High	Adequate	Loud	Safe	Suspicious	Expert	5.55	Overwhelming	Scarce	Heavy	0.98
628	High	Adequate	Loud	Safe	Spotted	Novice	6.05	Overwhelming	Scarce	Heavy	1.00
629	High	Adequate	Loud	Safe	Spotted	Average	6.65	Overwhelming	Scarce	Heavy	1.00
630	High	Adequate	Loud	Safe	Spotted	Expert	7.25	Overwhelming	Scarce	Heavy	1.00
631	High	Adequate	Loud	Engaged	Hidden	Novice	2.05	Passive	Scarce	Mixed	0.70
632	High	Adequate	Loud	Engaged	Hidden	Average	2.65	Tactical	Scarce	Mixed	0.74
633	High	Adequate	Loud	Engaged	Hidden	Expert	3.25	Tactical	Scarce	Heavy	0.98
634	High	Adequate	Loud	Engaged	Suspicious	Novice	3.75	Tactical	Scarce	Mixed	0.94
635	High	Adequate	Loud	Engaged	Suspicious	Average	4.35	Tactical	Scarce	Heavy	0.70
636	High	Adequate	Loud	Engaged	Suspicious	Expert	4.95	Overwhelming	Scarce	Heavy	0.74
637	High	Adequate	Loud	Engaged	Spotted	Novice	5.45	Overwhelming	Scarce	Heavy	0.94
638	High	Adequate	Loud	Engaged	Spotted	Average	6.05	Overwhelming	Scarce	Heavy	1.00
639	High	Adequate	Loud	Engaged	Spotted	Expert	6.65	Overwhelming	Scarce	Heavy	1.00
640	High	Adequate	Loud	Swarmed	Hidden	Novice	1.45	Passive	Balanced	Mixed	0.94
641	High	Adequate	Loud	Swarmed	Hidden	Average	2.05	Passive	Balanced	Mixed	0.70
642	High	Adequate	Loud	Swarmed	Hidden	Expert	2.65	Tactical	Balanced	Heavy	0.74
643	High	Adequate	Loud	Swarmed	Suspicious	Novice	3.15	Tactical	Balanced	Mixed	0.94
644	High	Adequate	Loud	Swarmed	Suspicious	Average	3.75	Tactical	Balanced	Heavy	0.94
645	High	Adequate	Loud	Swarmed	Suspicious	Expert	4.35	Tactical	Balanced	Heavy	0.70
646	High	Adequate	Loud	Swarmed	Spotted	Novice	4.85	Overwhelming	Balanced	Heavy	0.70
647	High	Adequate	Loud	Swarmed	Spotted	Average	5.45	Overwhelming	Balanced	Heavy	0.94
648	High	Adequate	Loud	Swarmed	Spotted	Expert	6.05	Overwhelming	Balanced	Heavy	1.00

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
649	High	Surplus	Quiet	Safe	Hidden	Novice	1.60	Passive	Scarce	Swarm	0.88
650	High	Surplus	Quiet	Safe	Hidden	Average	2.20	Passive	Scarce	Mixed	0.64
651	High	Surplus	Quiet	Safe	Hidden	Expert	2.80	Tactical	Scarce	Mixed	0.80
652	High	Surplus	Quiet	Safe	Suspicious	Novice	3.30	Tactical	Scarce	Mixed	1.00
653	High	Surplus	Quiet	Safe	Suspicious	Average	3.90	Tactical	Scarce	Mixed	0.88
654	High	Surplus	Quiet	Safe	Suspicious	Expert	4.50	Tactical	Scarce	Mixed	0.64
655	High	Surplus	Quiet	Safe	Spotted	Novice	5.00	Overwhelming	Scarce	Mixed	0.76
656	High	Surplus	Quiet	Safe	Spotted	Average	5.60	Overwhelming	Scarce	Heavy	1.00
657	High	Surplus	Quiet	Safe	Spotted	Expert	6.20	Overwhelming	Scarce	Heavy	1.00
658	High	Surplus	Quiet	Engaged	Hidden	Novice	1.00	Passive	Scarce	Swarm	1.00
659	High	Surplus	Quiet	Engaged	Hidden	Average	1.60	Passive	Scarce	Mixed	0.88
660	High	Surplus	Quiet	Engaged	Hidden	Expert	2.20	Passive	Scarce	Mixed	0.64
661	High	Surplus	Quiet	Engaged	Suspicious	Novice	2.70	Tactical	Scarce	Mixed	0.76
662	High	Surplus	Quiet	Engaged	Suspicious	Average	3.30	Tactical	Scarce	Mixed	1.00
663	High	Surplus	Quiet	Engaged	Suspicious	Expert	3.90	Tactical	Scarce	Mixed	0.88
664	High	Surplus	Quiet	Engaged	Spotted	Novice	4.40	Tactical	Scarce	Mixed	0.68
665	High	Surplus	Quiet	Engaged	Spotted	Average	5.00	Overwhelming	Scarce	Heavy	0.76
666	High	Surplus	Quiet	Engaged	Spotted	Expert	5.60	Overwhelming	Scarce	Heavy	1.00
667	High	Surplus	Quiet	Swarmed	Hidden	Novice	0.40	Passive	Scarce	Swarm	1.00
668	High	Surplus	Quiet	Swarmed	Hidden	Average	1.00	Passive	Scarce	Mixed	1.00
669	High	Surplus	Quiet	Swarmed	Hidden	Expert	1.60	Passive	Scarce	Mixed	0.88
670	High	Surplus	Quiet	Swarmed	Suspicious	Novice	2.10	Passive	Scarce	Mixed	0.68
671	High	Surplus	Quiet	Swarmed	Suspicious	Average	2.70	Tactical	Scarce	Mixed	0.76
672	High	Surplus	Quiet	Swarmed	Suspicious	Expert	3.30	Tactical	Scarce	Mixed	1.00
673	High	Surplus	Quiet	Swarmed	Spotted	Novice	3.80	Tactical	Scarce	Mixed	0.92
674	High	Surplus	Quiet	Swarmed	Spotted	Average	4.40	Tactical	Scarce	Heavy	0.68
675	High	Surplus	Quiet	Swarmed	Spotted	Expert	5.00	Overwhelming	Scarce	Heavy	0.76
676	High	Surplus	Moderate	Safe	Hidden	Novice	2.40	Tactical	Scarce	Mixed	0.64
677	High	Surplus	Moderate	Safe	Hidden	Average	3.00	Tactical	Scarce	Mixed	0.88
678	High	Surplus	Moderate	Safe	Hidden	Expert	3.60	Tactical	Scarce	Mixed	1.00
679	High	Surplus	Moderate	Safe	Suspicious	Novice	4.10	Tactical	Scarce	Mixed	0.80
680	High	Surplus	Moderate	Safe	Suspicious	Average	4.70	Overwhelming	Scarce	Heavy	0.64
681	High	Surplus	Moderate	Safe	Suspicious	Expert	5.30	Overwhelming	Scarce	Heavy	0.88
682	High	Surplus	Moderate	Safe	Spotted	Novice	5.80	Overwhelming	Scarce	Heavy	1.00
683	High	Surplus	Moderate	Safe	Spotted	Average	6.40	Overwhelming	Scarce	Heavy	1.00
684	High	Surplus	Moderate	Safe	Spotted	Expert	7.00	Overwhelming	Scarce	Heavy	1.00
685	High	Surplus	Moderate	Engaged	Hidden	Novice	1.80	Passive	Scarce	Mixed	0.80
686	High	Surplus	Moderate	Engaged	Hidden	Average	2.40	Tactical	Scarce	Mixed	0.64
687	High	Surplus	Moderate	Engaged	Hidden	Expert	3.00	Tactical	Scarce	Mixed	0.88
688	High	Surplus	Moderate	Engaged	Suspicious	Novice	3.50	Tactical	Scarce	Mixed	1.00
689	High	Surplus	Moderate	Engaged	Suspicious	Average	4.10	Tactical	Scarce	Heavy	0.80
690	High	Surplus	Moderate	Engaged	Suspicious	Expert	4.70	Overwhelming	Scarce	Heavy	0.64
691	High	Surplus	Moderate	Engaged	Spotted	Novice	5.20	Overwhelming	Scarce	Heavy	0.84
692	High	Surplus	Moderate	Engaged	Spotted	Average	5.80	Overwhelming	Scarce	Heavy	1.00
693	High	Surplus	Moderate	Engaged	Spotted	Expert	6.40	Overwhelming	Scarce	Heavy	1.00
694	High	Surplus	Moderate	Swarmed	Hidden	Novice	1.20	Passive	Scarce	Mixed	1.00
695	High	Surplus	Moderate	Swarmed	Hidden	Average	1.80	Passive	Scarce	Mixed	0.80

#	Health	Ammo	Noise	Pressure	Exposure	Skill	Score	THEN Threat	THEN Supply	THEN Compo	w
696	High	Surplus	Moderate	Swarmed	Hidden	Expert	2.40	Tactical	Scarce	Mixed	0.64
697	High	Surplus	Moderate	Swarmed	Suspicious	Novice	2.90	Tactical	Scarce	Mixed	0.84
698	High	Surplus	Moderate	Swarmed	Suspicious	Average	3.50	Tactical	Scarce	Heavy	1.00
699	High	Surplus	Moderate	Swarmed	Suspicious	Expert	4.10	Tactical	Scarce	Heavy	0.80
700	High	Surplus	Moderate	Swarmed	Spotted	Novice	4.60	Tactical	Scarce	Heavy	0.60
701	High	Surplus	Moderate	Swarmed	Spotted	Average	5.20	Overwhelming	Scarce	Heavy	0.84
702	High	Surplus	Moderate	Swarmed	Spotted	Expert	5.80	Overwhelming	Scarce	Heavy	1.00
703	High	Surplus	Loud	Safe	Hidden	Novice	3.20	Tactical	Scarce	Mixed	0.96
704	High	Surplus	Loud	Safe	Hidden	Average	3.80	Tactical	Scarce	Heavy	0.92
705	High	Surplus	Loud	Safe	Hidden	Expert	4.40	Tactical	Scarce	Heavy	0.68
706	High	Surplus	Loud	Safe	Suspicious	Novice	4.90	Overwhelming	Scarce	Heavy	0.72
707	High	Surplus	Loud	Safe	Suspicious	Average	5.50	Overwhelming	Scarce	Heavy	0.96
708	High	Surplus	Loud	Safe	Suspicious	Expert	6.10	Overwhelming	Scarce	Heavy	1.00
709	High	Surplus	Loud	Safe	Spotted	Novice	6.60	Overwhelming	Scarce	Heavy	1.00
710	High	Surplus	Loud	Safe	Spotted	Average	7.20	Overwhelming	Scarce	Heavy	1.00
711	High	Surplus	Loud	Safe	Spotted	Expert	7.80	Overwhelming	Scarce	Heavy	1.00
712	High	Surplus	Loud	Engaged	Hidden	Novice	2.60	Tactical	Scarce	Mixed	0.72
713	High	Surplus	Loud	Engaged	Hidden	Average	3.20	Tactical	Scarce	Heavy	0.96
714	High	Surplus	Loud	Engaged	Hidden	Expert	3.80	Tactical	Scarce	Heavy	0.92
715	High	Surplus	Loud	Engaged	Suspicious	Novice	4.30	Tactical	Scarce	Heavy	0.72
716	High	Surplus	Loud	Engaged	Suspicious	Average	4.90	Overwhelming	Scarce	Heavy	0.72
717	High	Surplus	Loud	Engaged	Suspicious	Expert	5.50	Overwhelming	Scarce	Heavy	0.96
718	High	Surplus	Loud	Engaged	Spotted	Novice	6.00	Overwhelming	Scarce	Heavy	1.00
719	High	Surplus	Loud	Engaged	Spotted	Average	6.60	Overwhelming	Scarce	Heavy	1.00
720	High	Surplus	Loud	Engaged	Spotted	Expert	7.20	Overwhelming	Scarce	Heavy	1.00
721	High	Surplus	Loud	Swarmed	Hidden	Novice	2.00	Passive	Scarce	Mixed	0.72
722	High	Surplus	Loud	Swarmed	Hidden	Average	2.60	Tactical	Scarce	Heavy	0.72
723	High	Surplus	Loud	Swarmed	Hidden	Expert	3.20	Tactical	Scarce	Heavy	0.96
724	High	Surplus	Loud	Swarmed	Suspicious	Novice	3.70	Tactical	Scarce	Heavy	0.96
725	High	Surplus	Loud	Swarmed	Suspicious	Average	4.30	Tactical	Scarce	Heavy	0.72
726	High	Surplus	Loud	Swarmed	Suspicious	Expert	4.90	Overwhelming	Scarce	Heavy	0.72
727	High	Surplus	Loud	Swarmed	Spotted	Novice	5.40	Overwhelming	Scarce	Heavy	0.92
728	High	Surplus	Loud	Swarmed	Spotted	Average	6.00	Overwhelming	Scarce	Heavy	1.00
729	High	Surplus	Loud	Swarmed	Spotted	Expert	6.60	Overwhelming	Scarce	Heavy	1.00